



Regular Manuscript

Website cookies and voluntary disclosure[☆]Junhao Liu^{ID}

Faculty of Business and Economics, University of Melbourne, Australia



ARTICLE INFO

JEL classification:

G14
L86
M15
M40
M41
O33

Keywords:

Website cookies
Voluntary disclosure
Digital economy
Financial reporting
Internal information quality
Big data
Data analytics
Data privacy

ABSTRACT

Using website cookie information from U.S. firms' websites, I measure consumer data collected through cookies and examine how such data affect firms' accounting information environments. Cookies provide granular and real-time consumer data to firms' internal information environments and may improve the quality of internal information that managers rely on for external reporting. I find that firms with more cookies issue management sales forecasts more frequently and devote a larger share of disclosures in 10-K filings to customer, marketing, and product topics. The effects are stronger when cookie-collected data are more relevant or larger in volume and when firms have better data analytic technology. However, proprietary costs and data privacy protection mechanisms attenuate these effects. Using the California Consumer Privacy Act as a quasi-natural experiment, I provide causal evidence linking cookie-collected data to voluntary disclosure. Overall, the study highlights the role of firm-collected consumer data in shaping accounting information environments.

1. Introduction

Consumer data have become a key driver of economic growth and corporate decision-making (Brynjolfsson and McElheran, 2016; Goldfarb and Tucker, 2019; Goldfarb and Que, 2023; Veldkamp, 2023; Kepler et al., 2025). Firms collect large volumes of data directly from consumers through websites, mobile apps, and other digital tools to support business activities such as customer management, targeted advertising, product analysis, and demand prediction (Wedel and Kannan, 2016; Cohen, 2018; Tong et al., 2020). Beyond their operational role, consumer data may also be relevant for financial reporting because they convey information about sales performance, customer base, marketing effectiveness, and product popularity. The collection of granular and real-time consumer data into the internal information environment may influence the quality of internal information that managers rely on for financial reporting and thus affect external disclosure decisions (Verrecchia, 1990; Hemmer and Labro, 2008; Gallemore and Labro, 2015; Samuels, 2021). In the digital era, the influx of consumer data has the potential to shape the way managers acquire

[☆] I thank Wayne Guay (editor) and an anonymous reviewer for their thoughtful and constructive feedback. This paper is based on my dissertation at the Rotman School of Management, University of Toronto. I am extremely grateful to my dissertation committee – Ole-Kristian Hope (chair), Akash Chattopadhyay, Andrea Down, Avi Goldfarb, and Hai Lu – for continuous guidance and support. I also thank Mary Billings, Francesco Bova, Holger Daske (discussant), Xingchao Gao, Jody Grewal, Shuqing Huang, Scott Liao, Christine Liu, Hunter Pearson (discussant), Verina Que, Gord Richardson, Clark Shu, Xijiang Su, Jenny Tucker, Kevin Veenstra, Dushyant Vyas, Danye Wang, Baohua Xin, Mingyue Zhang, Christina Zhu, and seminar participants at New York University, Singapore Management University, Texas Christian University, the Frankfurt School of Finance and Management, the University of British Columbia, the University of Hong Kong, the University of Melbourne, the University of Toronto, the University of Washington, the 2023 Emerging Scholars in Accounting Conference, and the 2024 AAA Annual Meeting for valuable comments and suggestions. I gratefully acknowledge the financial support from the TD Management Data Analytics Lab. All errors are my own.

E-mail address: junhao.liu@unimelb.edu.au.

<https://doi.org/10.1016/j.jacceco.2026.101885>

Received 7 November 2023; Received in revised form 27 January 2026; Accepted 10 March 2026

Available online 19 March 2026

0165-4101/© 2026 The Author. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

and synthesize internal information for external financial reporting. To shed light on this issue, this paper focuses on the role of firm-collected consumer data in the internal and external information environments by examining the relation between these data and voluntary disclosures.

A major challenge for studying firm-collected consumer data lies in measurement, as these data are largely proprietary and unobservable. I address the challenge by exploiting *publicly* available information on *website cookies*, which are small programs and scripts used by websites to track and collect data from users. While it is infeasible to access the underlying data that firms collect through cookies, cookie information makes it possible to infer the volume and the content of these data.

Cookies are economically important. First, although cookies are not the only source of consumer data, cookie-collected data are fundamental to digital marketing and customer relationship management, allowing firms to characterize consumer preferences and behavior patterns, deliver personalized advertisements, analyze product demand, and ultimately enhance revenue and profitability (Goldfarb and Tucker, 2011a; Lambrecht and Tucker, 2013). Second, cookies are widely used due to their adaptability and relatively low technical requirements, with 90% of top-million websites using cookies (Munir et al., 2023). Third, cookies tend to capture data that are more granular, more timely, and broader in scope than traditional sources of consumer data. In turn, such data may provide incremental information that informs managers' assessments of historical performance and forecasts of future prospects.¹

Beyond their economic importance, cookies offer unique advantages for this study. First, publicly available cookie information enables the construction of a *generalizable* measure of firm-collected consumer data. Moreover, the variation in the quantity and composition of cookies can be used to capture differences in data collection practices across firms. Second, cookie usage is plausibly exogenous to firms' disclosure choices and the designs of their accounting information systems. As cookies are mainly used for marketing and advertising and are managed by sales or marketing teams, their adoption is unlikely to be driven by financial reporting or disclosure incentives.

This study focuses on whether cookie-collected consumer data affect voluntary disclosure decisions. Conceptually, this relation depends on (1) the extent to which cookie-collected consumer data improve the quality of internal information and (2) managers' incentives to disclose the incremental information learned from cookie data. Accordingly, I develop the hypothesis in two steps. First, I argue that cookies improve the quality of internal information relevant for external reporting by supplying large volumes of granular and real-time data on sales-related activities and customer behavioral patterns. Second, I expect that managers with better internal information from cookie-collected data are more likely to issue voluntary disclosures regarding these aspects, as high-quality internal information lowers disclosure thresholds, thus encouraging disclosures (Dye, 1985; Jung and Kwon, 1988; Verrecchia, 1990; Hemmer and Labro, 2008).

The hypothesis is not without tension. First, cookies do not generate accounting information but contain signals about consumer behaviors and sales-related activities. Such signals may not be translated into useful information for financial reporting if they are limited in volume, difficult to analyze due to firms' lack of data analytic capabilities, or constrained by data privacy protection. Second, even if cookie data enhance internal information quality, managers may choose to withhold disclosure due to proprietary costs (Verrecchia, 1983; Darrrough and Stoughton, 1990). Proprietary costs are particularly salient in this context because cookie-collected data relate directly to core operations and consumer activities. In addition, data privacy regulations and reputational concerns may further discourage disclosure based on cookie-collected consumer data, as firms may fear regulatory scrutiny and consumer backlash associated with the collection and use of consumer data (Martin and Murphy, 2017). Consequently, it remains an open question *ex ante* whether and how cookie-collected data affect voluntary disclosures.

To construct a measure of consumer data collected by cookies, I employ OpenWPM to collect the metadata on cookies from firms' *consumer-oriented* websites. OpenWPM simulates human interactions with websites and records information on each cookie file (Englehardt and Narayanan, 2016; Englehardt, 2018). As cookie information can only be obtained through live interactions with websites, it is technically impossible to collect historical cookie metadata.² Therefore, the primary analyses rely on a cross-sectional sample with cookie information collected from U.S. public firms' websites in February 2023, covering 2860 unique firms and containing 151,512 cookie files.

I measure the volume of cookie-collected data using the number of cookies, a measure that is positively associated with the number of variables collected per user.³ I further exploit cookie characteristics – ownership, purpose, and duration – to identify the variation in the volume and relevance of cookie-collected data across firms.

I employ two sets of voluntary disclosure proxies that align with the nature of cookie-collected consumer data. First, following Samuels (2021), I use the frequency and the likelihood of management sales forecasts, which capture the role of cookie-collected data in forecasting future purchases based on consumer preferences and behavioral patterns. Second, as cookie-collected data may also inform qualitative discussions regarding customers, marketing, and products, I employ the FinBERT model developed by Yang et al. (2020) and Huang et al. (2023) to measure the proportion of qualitative discussions on these topics in 10-K filings. To supplement the proxies capturing the *quantity* of voluntary disclosure, I also consider proxies for the *quality* of voluntary disclosure, including the precision and the accuracy of management sales forecasts (Feng et al., 2009; Ittner and Michels, 2017).

In the baseline analyses, the number of cookies is positively associated with the frequency and the likelihood of management sales forecasts, as well as the percentage of sentences in 10-K filings discussing customers, marketing, and products. These results suggest

¹ See Section 2 for more detailed discussions.

² Static snapshots of websites (e.g., the Wayback Machine) do not contain cookie information and cannot be used to extract cookie information.

³ Note that cookies are different from website traffic used by Armstrong et al. (2025). If cookie-collected data are stored in a data table, cookies specify the variables to be collected (i.e., the columns of the data table), whereas website traffic captures the number of observations (i.e., the rows of a data table).

that firms collecting more consumer data through cookies tend to issue more forecasts and disclose more information about their sales performance and customers. The findings are robust to controlling for different sets of firm and industry characteristics, data analytic technology proxies, and industry fixed effects. In addition, the impact threshold of confounding variables (ITCV) analysis (e.g., [Larcker and Rusticus, 2010](#)) suggests that it is unlikely that omitted variable bias would overturn the inferences.

Building on the baseline results, I evaluate factors affecting the extent to which cookie-collected consumer data enhance internal information quality. First, I examine the role of data relevance and volume. Cookies differ in ownership, purpose, and duration, and these characteristics are closely related to the relevance and the volume of the data collected. Exploiting cross-sectional variation in cookie composition across firms, I find that the baseline effect is stronger when firms use more cookies that collect data of greater relevance to financial reporting (i.e., first-party cookies and analytic cookies) or that collect a larger volume of data (i.e., persistent cookies). The results validate the first conceptual link underlying the main hypothesis and reinforce the baseline findings.

Next, the usefulness of cookie-collected data depends not only on the data but also on firms' ability to manage and analyze them effectively. As cookie-collected data are largely semi-structured or unstructured, their informational value hinges on firms' data analytic capabilities ([LaValle et al., 2010](#); [Kiron et al., 2013](#)). Using the Aberdeen CI Technology Database to construct proxies, I evaluate the role of different facets of data analytic technology, including hardware capacity, software availability, IT investments, and human capital. The results show that the positive effect of cookies on disclosure is stronger for firms with stronger data analytic technology, consistent with a complementary relation between data and analytics.

I then evaluate factors that may affect managers' incentives to disclose the information learned from cookie-collected data. As cookie-collected data contain proprietary information, such as granular sales performance and customer activities, managers may withhold disclosures when the proprietary costs of disclosing such information are high. To test this possibility, I examine whether the effect of cookie-collected data on disclosure varies with industry competition. Using the Herfindahl–Hirschman Index (HHI) of sales revenue at the GICS four-digit level, I find that the relation between cookies and disclosure is weaker in industries with more intense competition, suggesting that managers may withhold the information learned from cookies when proprietary costs are high.

Finally, I examine the role of data privacy protection. Data privacy protection may affect the relation between cookie-collected data and voluntary disclosure through two channels. First, data privacy protection may restrict firms' data collection practices and reduce the effectiveness of cookie-collected data in improving internal information quality ([Jia et al., 2021](#); [Goldberg et al., 2024](#)). Second, public awareness of firms' data collection and their data privacy rights may deter firms from disclosing the insights learned from cookie-collected data ([Goldfarb and Que, 2023](#)). To assess the effects of data privacy protection, I first examine a mandatory regulation, the California Consumer Privacy Act (CCPA). Using the introduction of the CCPA in 2020 in a difference-in-differences (DID) design, I show that a plausibly exogenous restriction on data collection reduces voluntary disclosures among treatment firms. The test complements the primary analyses by extending the finding to panel data and strengthening causal inferences.

I next evaluate the voluntary adoption of cookie consent banners. These banners disclose cookie usage and provide opt-out options for users, thus limiting firms' collection of consumer data. I show that cookies are less useful when cookie banners are in place, although the effect for firms that provide both disclosure and opt-out options is not significantly stronger than that for firms that merely disclose cookie usage. The insignificant result may be driven by self-selection, as firms offering opt-out options may have better technology to offset the loss of data or be able to infer consumers' demographics and preferences from their opt-out choices ([Johnson et al., 2020](#)).

I conduct several additional analyses to further strengthen the inferences of the paper. First, I show that cookies not only increase the quantity of voluntary disclosures but also improve their quality, as measured by the precision and the accuracy of management sales forecasts. Second, the effects are stronger for consumer-oriented industries, where cookies and consumer data are more important, but weaker for larger firms and IT firms, which likely have access to alternative data collection tools and rely less on cookies.

My study contributes to research and practice in several ways. First, it extends the literature linking internal information quality and external reporting ([Dye, 1985](#); [Jung and Kwon, 1988](#); [Verrecchia, 1990](#); [Hemmer and Labro, 2008](#)). Conceptually, internal information quality depends not only on firms' information processing systems but also on data inputs ([Neely and Cook, 2011](#)). Prior studies primarily emphasize the role of internal systems, such as ERP adoption and internal controls, in improving internal information quality and external disclosures ([Feng et al., 2009](#); [Dorantes et al., 2013](#); [Gallemore and Labro, 2015](#); [Ittner and Michels, 2017](#); [Samuels, 2021](#)). This paper differs by explicitly focusing on the role of data, alongside the internal systems and processes, in shaping the internal information environment. As digitalization expands, firms experience rapid growth in consumer data without commensurate changes in accounting information systems or reporting infrastructure. As a result, variation in internal information quality can arise not only from internal systems and processes but also from the influx of underlying consumer data. The paper's contribution is to explicitly identify data as a factor affecting internal information quality and to empirically assess the role of firm-collected consumer data in the corporate information environment.

Second, the paper addresses the empirical challenge of measuring firms' consumer data collection activities and the data collected. Although consumer data are increasingly recognized as a key category of intangible capital (e.g., [Kepler et al., 2025](#)), their proprietary and unobservable nature makes them difficult to measure. Recent studies have made important progress by exploiting the launch of mobile apps ([Ferracuti et al., 2024](#)) and purchase price allocations ([Kepler et al., 2025](#)). However, these approaches may not generalize broadly or capture rich variation in data practices. This paper introduces a complementary approach by leveraging publicly available information on a widely used data collection tool (i.e., website cookies) to infer and quantify the volume and the content of firm-collected consumer data. This approach may open new avenues for empirical research on firm-level data collection practices.

Finally, this paper contributes to the debate on corporate data collection and consumer data privacy protection. Research suggests that corporate data collection benefits firms' marketing and advertising activities but harms consumers' data privacy (Goldfarb and Tucker, 2011a, 2012; Martin and Murphy, 2017; Bian et al., 2025; Goldfarb and Que, 2023). This study takes a different perspective and implies a potential benefit of firms' collection and usage of consumer data: such data may be useful in enhancing the transparency of the corporate information environment. The paper may inform the debates on the social costs and benefits of restricting corporate data collection.

2. Background

2.1. Website cookies

Amid the rapid development of the digital economy, firms increasingly use digital platforms and tools to accumulate data from stakeholders, especially individual consumers. These data have become important drivers of productivity and play significant roles in corporate decision-making and value creation (Brynjolfsson and McElheran, 2016; Goldfarb and Tucker, 2019; Veldkamp, 2023; Kepler et al., 2025).

Website cookies are one of the most widely used digital data collection tools and are designed to acquire large-scale, granular, and real-time data from website users (Goldfarb and Tucker, 2011a). Cookies are small text files or scripts that are sent from website servers to users' browsers and collect data from them. As of 2023, nearly 90% of top-million websites use cookies to track their users (Munir et al., 2023). Compared with other data collection tools, such as mobile apps, cookies are more widely used because they are less costly and have relatively low technological requirements.⁴ Despite relatively low adoption costs, cookies are powerful in tracking and collecting data from users across different websites and platforms.

As an essential component of websites, cookies were initially designed in the 1990s to create a reliable mechanism for websites to record user-specific information and perform useful functions, such as remembering user credentials, maintaining browsing sessions, and detecting automated traffic. With the development of the Internet, cookies have been increasingly used to track and collect data from users.⁵ Firms analyze these cookie-collected data to identify consumer preferences and behavioral patterns, deliver personalized advertisements, analyze product popularity, estimate purchasing intent and spending power, and ultimately improve marketing efficiency and profitability (Goldfarb and Tucker, 2011a; Lambrecht and Tucker, 2013; Goldfarb and Tucker, 2019).

Cookie-collected data are likely to provide managers with incremental information about sales operations and customer bases. There are several reasons. First, cookie-collected data are more granular than those obtained through traditional sources, such as marketing intelligence reports and consumer surveys. Cookies allow firms not only to collect data from each individual consumer for each visit in detail but also to aggregate these data at multiple levels. Second, cookie-collected data are timely and allow managers to obtain real-time information. For example, cookie-collected data allow executives to monitor sales and consumer activities in real time without relying on periodic reports from individual business units. Third, cookies offer data of a broader scope on the customer base, covering not only existing customers but also potential customers who browse the website without purchasing. Such information is unavailable from transaction records or point-of-sale systems but can be useful for predicting future sales and understanding customer retention and attrition.⁶

In this study, I exploit publicly available cookie information to infer and measure unobservable firm-collected consumer data. More details are provided in Section 4.1.

2.2. Data privacy protection

For years, website owners have gathered large volumes of user data without notice or consent, and some firms have even abused cookies to track users and resell personal data (Englehardt and Narayanan, 2016; Englehardt, 2018). Intensive cookie usage and corporate data collection practices have raised concerns about consumer data privacy and attracted public and regulatory scrutiny (Wang et al., 1998; Goldfarb and Tucker, 2011b, 2012; Jia et al., 2021; Goldfarb and Que, 2023; Goldberg et al., 2024).

In response to such concerns, legislators have introduced *mandatory* data privacy laws to restrict corporate data collection. In the U.S., California led with the California Consumer Privacy Act (CCPA), passed in 2018 and effective in 2020. In addition, some U.S. firms have *voluntarily* adopted cookie consent banners to inform users about data collection.⁷ These banners are used to (1) disclose the usage of cookies for data collection and (2) allow users to customize the types of cookies and block unnecessary cookies.

⁴ Ferracuti et al. (2024) find that around 30% of U.S. firms have mobile apps as of 2021. The percentage is lower than that of firms using cookies (90%, 3347 out of 3717 firms with valid URLs in the sample).

⁵ Cookies can collect a wide range of user-specific data that allow websites and firms to track, classify, and target individuals online. These data include but are not limited to (1) demographic data on user characteristics such as age, gender, language, and geographic locations, (2) behavioral data that record how users interact with websites (e.g., pages visited, time spent on each page, navigation paths, and click activities), and (3) conversion data about users' source platforms and whether website visits result in purchases or other desired actions.

⁶ Even for firms without e-commerce operations, browsing data may provide early indicators of offline demand, as consumers often research products or store locations online before visiting physical outlets. The geographic and demographic patterns inferred from cookies can help firms anticipate regional sales trends and allocate marketing resources more effectively.

⁷ For most U.S. websites, disclosing the usage of cookies is *voluntary*. In Europe, cookie consent banners are mandatory due to the requirements of the General Data Protection Regulation (GDPR).

Such data privacy protection initiatives may restrict firms' data collection practices and adversely affect the volume and the quality of cookie-collected data (Jia et al., 2021; Goldberg et al., 2024). In addition, data privacy protection initiatives may raise public awareness of firms' data collection activities, increasing firms' reputational concerns when making use of collected consumer data in their business operations or disclosures (Martin and Murphy, 2017). As a result, data privacy protection may reduce the effectiveness of cookies in improving the information environment.

3. Hypothesis development

I examine whether and how cookie-collected consumer data shape firms' internal information quality and, in turn, influence voluntary disclosures. This study is related to prior studies linking internal information quality and external reporting (e.g., Feng et al., 2009; Dorantes et al., 2013; Samuels, 2021) but focuses on a different source of internal information quality. Conceptually, internal information quality is a function of (1) data inputs and (2) the internal systems processing those inputs (Neely and Cook, 2011). Prior studies primarily focus on the latter, examining how internal information systems and processes affect internal information quality.⁸ This study extends the literature by explicitly identifying and measuring the underlying data inputs that feed these internal information systems. As firms experience exponential growth in data volume, granularity, and timeliness without commensurate changes in accounting information systems and processes, I expect that data inputs are becoming increasingly important for internal information quality.

I develop the hypothesis by discussing two links underlying the relation between cookie-collected consumer data and voluntary disclosure: (1) the extent to which cookie-collected consumer data improve internal information quality, and (2) managers' incentives to disclose conditional on learning incremental information from such data.

First, I argue that cookie-collected data improve the quality of internal information. As discussed in Section 2, cookies collect data from a large number of individual consumers and provide information about sales operations and customer bases. These data are likely to be incremental to traditional sources of consumer data due to their granularity, timeliness, and broad coverage. A potential concern is that cookie-collected data are primarily used by sales teams for digital marketing and may not be available to personnel responsible for financial reporting. However, given the integration of internal information systems, this concern is unlikely to be binding (Gallemore and Labro, 2015).⁹

As a necessary condition for cookie-collected data to be useful for disclosure, I explain why these data provide information relevant to financial reporting. Granular and real-time data on sales activities and consumer behavior patterns make it possible for firms to form more timely and accurate expectations about future sales, thus informing management guidance and other forward-looking disclosures. In addition, firms can exploit cookie-collected data to evaluate customer relationship management, marketing effectiveness, and product performance in detail, thus facilitating managers' qualitative discussions on these areas in the MD&A section.

Next, I discuss whether improved internal information quality translates into external disclosures. The internal and external information environments are interconnected, and changes in the internal information environment can manifest themselves in changes in external reporting (Hemmer and Labro, 2008). By using cookies, managers become better informed about sales operations and customers and receive more precise private signals. Disclosure theory predicts that a higher likelihood of being informed and greater precision of managers' private signals reduce disclosure thresholds and increase the likelihood of voluntary disclosure (Dye, 1985; Jung and Kwon, 1988; Verrecchia, 1990). In line with the theory, I predict that cookie-collected data are positively associated with the frequency and the likelihood of voluntary disclosure related to sales operations and customers.¹⁰

In summary, combining the two theoretical links discussed above, I predict that cookie-collected consumer data improve the quality of internal information about customers and sales-related activities and, in turn, encourage managers to voluntarily disclose more information about these aspects. I state my primary hypothesis in the alternative form as follows:

H1: Corporate cookie-collected data are positively associated with voluntary disclosure regarding customers and sales-related activities.

The effect of cookie-collected data on disclosure is not mechanical or inevitable. The paper's tension arises from two aspects: (1) whether and to what extent cookie-collected data improve the quality of internal information relevant for financial reporting, and (2) managers' disclosure incentives conditional on learning information from cookies.

First, cookies do not directly generate accounting signals but primarily capture consumer data for marketing and advertising purposes. While these data can inform managers about sales performance and customer engagement, they often provide noisy and probabilistic signals rather than realizations of sales or cash flows. Such data may not be converted into information useful for financial reporting if they are irrelevant for financial reporting or limited in volume. As these data are largely unstructured, the

⁸ For example, enterprise systems, such as ERP systems, improve the ability of firms to manage and process information, resulting in higher internal information quality and thus improved external disclosures (Dorantes et al., 2013). The internal control systems studied by Feng et al. (2009) are another category of internal processes that allow firms to monitor and improve internal information quality. Similarly, Samuels (2021) studies a setting in which government procurement contracts impose additional requirements on internal information processes and finds that improved internal information processes facilitate external reporting.

⁹ According to two anonymous experts in digital marketing, sales teams (or specialized digital marketing teams) are in charge of managing cookies and analyzing cookie-collected data. They communicate the insights from data analyses to top executives through executive memos and board meetings.

¹⁰ Under Dye (1985) and Jung and Kwon (1988), managers face a lower disclosure threshold when (1) they have a higher probability of being informed, and (2) outsiders have information on the probability that managers are informed. In my setting, the first condition is satisfied as cookies provide useful information to managers. The second condition holds to the extent that outsiders are aware that firms collect data from consumers.

effect may also be limited if firms do not have the necessary data analytic capabilities. Moreover, data privacy protection may diminish the usefulness of these data by constraining the permissible uses and reducing the volume of cookie-collected data.

Second, managers may refrain from disclosing insights learned from cookie-collected data when the costs of disclosure are high. As these data reveal sensitive information about sales and customer behavior, managers may withhold the information due to proprietary costs (Verrecchia, 1983; Darrough and Stoughton, 1990). In addition, data privacy protection initiatives may raise public awareness of firms' data collection practices and heighten managers' reputation concerns, thus deterring them from disclosing the information learned through cookies (Martin and Murphy, 2017; Goldberg et al., 2024).

I evaluate the countervailing forces discussed above in the empirical analyses to examine whether these forces strengthen or attenuate the role of cookies in disclosures.

4. Data and variables

4.1. Website cookies

4.1.1. Collecting cookie information

Cookies offer a practical approach to measuring firm-collected consumer data for a large sample of firms. While direct access to the underlying data collected by cookies is infeasible and illegal, publicly available cookie information reveals how firms collect consumer data and allow outsiders to infer the volume and content of these data.

I collect cookie information from company websites using OpenWPM (version 0.21.1, updated in February 2023), a Selenium-based automated browsing framework (Englehardt and Narayanan, 2016). Selenium automates web interactions such as clicking links and filling forms, and OpenWPM builds on it by simulating user behaviors in a full-fledged Firefox browser. For each cookie file received during browsing sessions, OpenWPM records detailed metadata, such as the name, value, expiration, and domain. Such information is then used to infer the volume and the types of data collected by cookies.

A key limitation of the study lies in the real-time nature of cookie data collection. It is technically infeasible to collect historical cookie data because cookies are generated dynamically through live interactions between website servers and browsers. As a result, the main analyses rely on cookies collected in February 2023.¹¹

I collect the website cookies of U.S. companies using the following steps. First, I start with U.S. publicly traded companies in the Compustat-CRSP merged data as of the end of 2022 and extract the website uniform resource locators (URLs) of these firms' consumer-oriented websites (4141 unique firms).¹² Website URLs for 47 firms are missing and cannot be manually identified. In addition, 397 observations belong to non-common stocks sharing the same websites, and these observations are either ADRs or closed-end funds. Many closed-end funds in the Compustat database share the same URLs of a few large asset management firms, and these observations are not meaningful in the context of this paper. Next, I use OpenWPM to obtain the cookies on each website. In this step, OpenWPM fails to collect cookie information from 460 websites due to website access issues such as HTTP 404 (website not found), HTTP 500 (internal server error), and HTTP 504 (gateway timeout). The process yields 151,512 cookies (20,532 unique cookies) from 3347 firms with valid URLs and successful visits.¹³

4.1.2. Classifying cookies

Cookies vary in ownership, purpose, and expiration. Cross-sectional differences in cookie composition across firms allow outsiders to identify variation in the relevance and the volume of cookie-collected data. Using cookie information collected by OpenWPM, I classify cookies into (1) first- and third-party cookies, (2) cookies with different purposes, and (3) session and persistent cookies.

First- and third-party cookies

A single website may host multiple cookies from different entities and domains. First-party cookies are managed by website owners and can be tailored to firms' data needs, whereas third-party cookies are controlled by external providers (e.g., Google, DoubleClick) and are primarily used for cross-site tracking and targeted advertising.¹⁴ As a result, compared with third-party cookies, first-party cookies are more likely to provide firm-specific information and are thus more relevant to the focal firm.

To classify cookies based on ownership, I compare the domain of each cookie with the domain of the website (e.g., Cheng et al., 2022). First-party cookies share or closely resemble the website domain (e.g., Zara's homepage <https://www.zara.com/> sets cookies from ".zara.com"), whereas third-party cookies originate from different domains, such as ".google.com" or ".doubleclick.net".

¹¹ Even archived website snapshots (e.g., the Wayback Machine) do not contain cookies. To strengthen the validity of the cookie data used in the study, I collect another snapshot of cookies in May 2023. In untabulated results, the two snapshots collected in February and May exhibit a positive correlation of 0.82, indicating some degree of time-series variation. About 58% of firms have made minor adjustments to their cookies (fewer than five cookies). 29% of firms have made changes to more than five cookies. The inferences of the paper are unaffected if I use the cookie data in May 2023 or the average of the two snapshots of data obtained in February and May (see Table IA1 in the Internet Appendix).

¹² Consumer-oriented pages refer to those webpages targeting consumers (e.g., e-commerce pages or product and service introduction pages). If consumer-oriented webpages are not available, I use the URL from the Compustat database instead.

¹³ Many cookies are from the same cookie service provider. Such cookies may appear on different websites. This explains why the number of unique cookies is smaller than the total number of cookies collected.

¹⁴ For example, a third-party cookie may record a user's activity on Website A and then enable Website B, which uses the same third-party cookie, to recognize the user and deliver personalized content based on the user's prior activity on Website A.

I conduct a fuzzy match between cookie and website domains. For each website, I calculate the Levenshtein similarity score between the website domain and each of the cookie domains (Levenshtein, 1966). I use 0.8 as a cutoff and classify cookies as first-party (third-party) cookies if the domain similarity is above (below) 0.8.¹⁵

Cookies with different purposes

Cookies serve different purposes. According to the GDPR, cookies can be classified as necessary, functional, analytic (or performance), and advertising (or targeting) cookies.¹⁶ Analytic cookies collect data on website usage and consumer behavior patterns. Such data may be more relevant to disclosure than the data collected for other purposes.¹⁷

To examine the role of cookies with different purposes in disclosure, I label the purposes of 151,512 cookies (20,532 unique cookies) in my sample by manually searching the Cookiepedia database (<https://cookiepedia.co.uk/>). Cookiepedia gathers most of its raw data from OneTrust, a major cookie consent management service provider. The Cookiepedia data allow users to check the likely purposes of cookies based on their names. It is possible to determine the purposes of cookies based on their names because many cookies are standardized and share similar naming patterns.¹⁸ This database has been widely used in academic research (e.g., Bollinger et al., 2022).

Session and persistent cookies

Each cookie has an expiration time, after which the cookie file is automatically deleted from the user's browser.¹⁹ Based on expiration time, cookies are classified as either session or persistent cookies. Session cookies are temporarily stored and deleted when the browser closes, while persistent cookies remain on the devices of users for a specified period, allowing firms to recognize returning visitors and collect time-series data.

Using the expiration time information, I classify cookies expiring within 24 h as session cookies and those lasting longer as persistent cookies. As persistent cookies track users across multiple visits and collect time-series data of each consumer, I expect them to collect larger volumes of data than session cookies.

4.2. Voluntary disclosure proxies

4.2.1. Management sales forecasts

Management forecasts are key informational events and are widely used as proxies for voluntary disclosure (Beyer et al., 2010). Following Samuels (2021), I employ management sales forecasts and use *frequency* and the *likelihood* of issuing management sales forecasts as my first set of voluntary disclosure proxies.²⁰

I primarily focus on management sales forecasts because sales forecasts are more closely related to cookie-collected consumer data compared to other management guidance.²¹ I use the I/B/E/S Guidance Database to construct the frequency of management sales forecasts in a year, as well as a binary variable that takes one if a firm issues at least one management sales forecast in a year and zero otherwise.

4.2.2. FinBERT-based measures

Cookies can provide managers with granular and real-time data about customers, marketing, and products. Managers may incorporate the insights learned from cookie-collected data into their qualitative discussions. Accordingly, I construct measures of qualitative disclosure about customers, marketing, and products at the sentence level. Specifically, I use the fine-tuned FinBERT model to classify sentences in Item 1 (Business), Item 1A (Risk Factors), and Item 7 (Management's Discussion and Analysis) in 10-K filings into sentences discussing customer-, marketing-, and product-related topics. Next, I calculate the percentage of sentences

¹⁵ My inferences are unaffected if I use alternative methods to calculate text similarity, such as cosine and Jaro-Winkler distances, or use different cutoffs, such as 0.6, 0.7, and 0.9.

¹⁶ The GDPR provides a widely accepted classification framework for cookies. Necessary cookies enable core website functions (e.g., session maintenance, login, shopping carts, bot detection). Users may not be able to browse the website without them. Functional cookies enhance the usability of websites and store users' settings (e.g., language preferences, social media sharing). Advertising or targeting cookies track users' online behaviors within one website or across different websites and platforms for targeted advertising.

¹⁷ Note that analytic cookies are not the only type of cookies collecting data useful for financial reporting. Other types of cookies may also collect certain personal data that overlap with the data collected by analytic cookies. For example, advertising cookies may rely on users' browsing history to generate personal recommendations. Functional cookies may provide useful information on consumers by capturing the input content from consumers.

¹⁸ Many websites do not create their own cookies from scratch but modify standardized cookies developed by cookie service companies. For example, "_ga" is one of the most commonly used cookies (with 6843 appearances in the sample of this paper). Google Analytics develops this cookie to assign unique IDs for each browser. This cookie can be used for analytics and may facilitate advertising and targeting. Some websites create their own non-standardized cookies, making it difficult to determine the purposes. However, the percentage is low (12.8% of the whole sample).

¹⁹ Users can manually delete cookies before expiration or use private browsing modes, which erase all cookies once the session ends.

²⁰ Focusing on the *quantity* of disclosure is consistent with disclosure theory and prior empirical studies (e.g., Samuels (2021)). In addition to the quantity of disclosure, it is possible that cookies are associated with the quality of management forecasts. Forecast accuracy and precision are two widely used proxies for the quality of management guidance (Feng et al., 2009; Dorantes et al., 2013; Ittner and Michels, 2017). I do not focus on management forecast precision in my primary analyses for two reasons. First, as suggested by Verrecchia (1990), precision is embedded in managers' decisions to issue voluntary disclosure. If the internal signal about future sales is not precise and reliable enough, managers may refrain from issuing forecasts because the costs of disclosure may outweigh the benefits. Hence, examining the precision of voluntary disclosure may introduce endogeneity and truncation problems. Second, focusing on precision largely reduces the sample size because the precision measure is available only if a firm issues management sales forecasts. Nonetheless, I use management sales forecast accuracy and precision as the dependent variables in Table 11 and find that cookies help improve management sales forecast accuracy and precision.

²¹ In the Internet Appendix, I also use management earnings forecasts and find consistent results (see Table IA2).

Table 1

Sample construction procedures.

This table tabulates sample construction procedures of this study. The sample is a cross-sectional sample constructed based on the website cookie information collected using OpenWPM in February 2023. The table describes all sample selection criteria and tabulates the number of observation lost from the sample and the number of remaining observations after each criterion is imposed.

Steps		Remaining
US firms in the Compustat-CRSP Merged Data of year 2022		4141
Less:		
Missing website URLs in Compustat and cannot be manually identified	(27)	4114
Non-common stocks sharing the same website	(397)	3,717
Websites cannot be accessed	(460)	3347
Observations with missing values in regression variables	(471)	2876
GICS four-digit industries with fewer than 10 firms	(16)	
Number of Observations for the sample		2,860

discussing each topic in each 10-K filing and use the percentages as the proxies for managers' qualitative disclosure on customer-, marketing-, and product-related topics. Please see [Appendix B](#) for a detailed discussion of model training and disclosure proxy construction.

4.3. Data analytic technology proxies

To evaluate how data analytic technology enables firms to utilize cookie-collected data in voluntary disclosure, I use information from the Aberdeen CI Technology Database (formerly Harte Hanks Market Intelligence). The database provides annual establishment-level data on firms' IT products and investments and has been widely used in economics, marketing, and information systems research (e.g., [Forman et al., 2005](#); [Bloom et al., 2012](#)). I exploit the Aberdeen data to measure different dimensions of firms' data analytic technology, including hardware capacity, software availability, and general data-analytic capability (IT human capital and budget). Details are available in Section 5.2.

4.4. Cookie banners

To understand the implications of data privacy protection for the usefulness of cookie-collected data in disclosure, I manually browse all websites to check the existence and the available options of cookie consent banners. 975 (34.8%) firms display cookie consent banners on their websites (i.e., disclose cookie usage to users). Among firms with banners, 356 firms provide "Decline All Cookies" or "Accept Necessary Cookies Only" options, allowing users to opt out of all unnecessary cookies with one click.²²

4.5. Other data

I employ the I/B/E/S Guidance Database to obtain data for disclosure proxies. To construct control variables, I use data from Compustat, CRSP, I/B/E/S, and Thomson Reuters 13F.

5. Research design

5.1. Sample construction

For the main analyses, I construct a cross-sectional sample using cookie information collected in February 2023. Starting from the 3347 firms with cookie information (as discussed in Section 4.1), I further exclude observations with missing values in regression variables (471 firms) and GICS four-digit industries with fewer than 10 firms (16 firms). [Table 1](#) describes the detailed procedures of sample selection. The final sample consists of 2860 unique firms.

5.2. Baseline model

For my primary analyses, I use a cross-sectional sample constructed using website cookie data collected in February 2023 and financial data in 2022.

To examine the effect of cookie-collected data on voluntary disclosure, I estimate the following regression model over the cross-sectional sample:

$$VolDisc = \beta_0 + \beta_1 Cookie + \sum_i \beta_i Control Variable_i + \varepsilon$$

²² See [Appendix A](#) for examples.

VolDisc denotes voluntary disclosure proxies, including the frequency of management sales forecasts *SF Freq*, the indicator for issuing a sales forecast, *SF*, and the proxies for qualitative disclosures about customers, marketing, and products in 10-K filings, as defined in Section 4.2. *Cookie* is the natural logarithm of the number of cookies on a firm's website, and the number of cookies is positively associated with the number of variables collected from each website user and thus the amount of consumer data collected (Englehardt and Narayanan, 2016).²³

Following prior studies (e.g., Samuels, 2021), I control for firm characteristics related to voluntary disclosure, including firm size (*Size*), market-to-book (*MB*), leverage (*Leverage*), return on equity (*ROE*), an indicator for losses (*Loss*), cash holdings (*Cash*), capital expenditures (*Capex*), asset tangibility (*PPE*), stock returns (*Return*), return volatility (*Volatility*), analyst coverage (*Analyst*), and institutional ownership (*Inst*).²⁴ I further include selling, general, and administrative expenses (*SG&A*) to account for firms' investments in marketing and advertising, which may be correlated with both cookie usage and disclosure practices. In addition, I control for internal information environment variables from Gallemore and Labro (2015), including report lag (*Report Lag*), internal control weaknesses (*ICW*), and an indicator for restatements (*Restate*).

As for industry characteristics, I include a binary variable for customer-oriented industries (*Customer*) to control for the possibility that consumer-oriented firms may use more cookies and issue more sales forecasts. I also use the Herfindahl–Hirschman index for sales concentration (*HHI*) to capture industry competition.²⁵

I employ the Aberdeen database to construct firm-level proxies for different aspects of data analytic technology. Hardware capacity is captured by data storage capacity (*Data Storage*) and the number of servers (*Server*). Software availability is measured using Aberdeen's scores (ranging from 0 to 5) for database management systems (*DBMS*), enterprise resource planning systems (*ERP*), and software-as-a-service applications (*SaaS*). I further measure IT-related human capital using the percentage of IT employees (*IT Employee*) and overall IT investment using IT budget as a share of sales revenue (*IT Budget*). These variables are relevant controls because data analytic technology may be correlated with both firms' cookie usage and their disclosure practices.

A limitation of the Aberdeen database is that it covers only about 40% of firms in the sample. Including all technology proxies would therefore substantially reduce the sample size. To mitigate this issue, I impute missing values as zero and include a missing-value indicator for each proxy (Donders et al., 2006).²⁶

Detailed definitions of all variables are available in Appendix C. All variables in this study are winsorized at the 1st and 99th percentiles.²⁷

In a cross-sectional sample, it is *infeasible* to control for firm or time fixed effects. Nonetheless, I include industry fixed effects (GICS four-digit industry groups) in one of the baseline specifications to control for industry-level time-invariant characteristics. To account for the possibility that observations are correlated within industries, I cluster standard errors at the GICS four-digit industry level.²⁸

5.3. DID analyses using the CCPA

Because the primary analyses are cross-sectional, unobserved firm- or industry-level factors may confound the results. To address this concern, I follow Cheng et al. (2022) and exploit the formal adoption of the California Consumer Privacy Act (CCPA) in 2020 using a difference-in-differences (DID) design.²⁹ Research suggests that data privacy regulation restricts companies' usage of cookies and their ability to collect consumer data (Jia et al., 2021; Goldberg et al., 2024). As the enactment and implementation of the CCPA are plausibly exogenous to firms' data collection and disclosure decisions, the CCPA provides a credible exogenous shock in this setting.

The CCPA applies to companies with (1) annual revenue above \$25 million, (2) data collection involving at least 50,000 California residents per year, or (3) at least 50% of revenue from selling personal data. It is challenging to obtain direct firm-level data on these criteria. To address this limitation, I focus on the relative strength of the treatment effects. Specifically, I first restrict the analyses to California firms to account for state-level confounding factors (e.g., legal environment, state-level regulation, etc.). Next,

²³ Inferences are robust to using the raw number counts in regressions.

²⁴ Note that including the number of analysts following the firm (*Analyst*) and institutional ownership (*Inst*) in the regression model may introduce multicollinearity problems and result in problematic estimates for firm size (*Size*). This is because these three variables are highly correlated within a cross-sectional sample – the correlation between *Size* and *Analyst* (*Inst*) is 0.67 (0.50). Also, as *Size*, *Analyst*, and *Inst* are all positively and significantly correlated with *Cookie*, it is likely that the standard error estimates for *Cookie* are inflated due to low statistical power. In such a sense, the regressions in the paper may underestimate the statistical significance of *Cookie*. In untabulated tests, I remove *Analyst* and *Inst* from the regression model, and the inferences throughout the paper are not affected.

²⁵ I define *Customer* and *HHI* based on GICS industry groups because this industry classification is designed based on firms' principal economic activities and revenue sources, which may be more relevant in distinguishing customer-oriented and non-customer-oriented firms (Bhojraj et al., 2003). Please see Appendix C for a detailed definition.

²⁶ Note that this approach may produce biased estimates for imputed variables if the observations are not missing at random (Donders et al., 2006). Nonetheless, the coefficient estimates for *Cookie* as well as other control variables remain quantitatively similar with and without these data analytic technology proxies.

²⁷ Inferences are unaffected by winsorization.

²⁸ The inferences are robust to using Huber-White robust standard errors or to not clustering. Note that clustering at the firm level is equivalent to no clustering for a cross-sectional sample.

²⁹ It is possible that firms preempt the formal adoption because the CCPA was passed in 2018. However, anticipatory effects bias against finding results in my setting.

I classify customer-oriented firms into the treatment group and non-customer-oriented firms into the control group. This is because customer-oriented firms are more likely to be affected by consumer privacy regulations.³⁰

I estimate the following regression model using all firms headquartered in California:

$$VolDisc_{i,t} = \beta_0 + \beta_1 CO_i \times Post\ CCPA_t + Control\ Variables + Industry\ or\ Firm\ FE_i + Year\ FE_t + \varepsilon_{i,t}$$

where CO_i takes one for firms in customer-oriented industries and zero otherwise. For both DID designs specified above, I control for the same set of variables as those in the baseline design and cluster standard errors at the firm level.

6. Results

6.1. Descriptive statistics

Panel A of Table 2 reports the industry distribution of cookies at the GICS four-digit level. Consistent with customer-oriented firms' heavier reliance on cookies, customer-oriented industries use more cookies on average (33.56) than non-customer-oriented industries (29.25). Retail firms use the most cookies (77.05), followed by transportation (62.23) and telecommunication services (58.91). The top five industries in the list are all customer-oriented. In contrast, pharmaceutical and energy firms use the fewest cookies, with an average of 14.36 and 12.32, respectively.

79.5% of cookies are set by first parties (i.e., firms). Customer-oriented firms use more first-party cookies than non-customer-oriented firms, implying that customer-oriented firms may invest more in first-hand data collection rather than relying on third parties. In terms of duration, 20.1% of cookies are session cookies that expire within 24 h, and 79.9% of cookies are persistent. By purpose, necessary, functional, analytic, and advertising cookies account for 16.0%, 3.9%, 18.2%, and 49.8%, respectively. These figures are consistent with those from Cookiepedia and Bollinger et al. (2022).

Panel B of Table 2 presents the descriptive statistics. 33% of firms issue at least one management sales forecast, and an average firm in the sample issues 1.12 sales forecasts. Customer-, marketing-, and product-related sentences account for 1.06%–1.88% of 10-K filings individually and about 4.6% in total.

Panel C shows the correlation matrix. Consistent with cookies facilitating voluntary disclosures, *Cookie* has significant and positive correlations with all the disclosure proxies. In addition, the use of cookies seems to be non-random but associated with firm characteristics, such as size, PP&E, SG&A, analyst coverage, institutional ownership, as well as industry characteristics.³¹ To account for selection based on these observed variables, I control for all these variables in the regressions.³²

Panel D tabulates the correlations with the proxies for data analytic technology.³³ *Cookie* still shows positive and significant correlations with the disclosure proxies. In addition, there are two important observations. First, *Cookie* is not significantly associated with firms' data analytic technology, implying that data collection is not the same as data analytics. It also indicates that using cookies on websites to collect data does not necessarily require advanced technology.³⁴ Second, data analytic technology proxies are not significantly correlated with disclosure proxies, reducing concerns that the observed relation between cookies and disclosure is driven by firms' analytics capabilities.

6.2. Baseline results: The usefulness of cookies for disclosure

I test whether cookie-collected consumer data enhance internal information quality and facilitate voluntary disclosure by regressing voluntary disclosure proxies on the number of cookies (*Cookie*). Table 3 tabulates the OLS regression results.³⁵ In Panel A, the dependent variable is the frequency of management sales forecasts. Column (1) includes only *Cookie* as the independent variable, and the coefficient is significant at the 1% level. Columns (2) to (4) further include control variables for firm and industry characteristics. The coefficient on *Cookie* declines but remains statistically significant. Columns (5) to (6) replace industry controls with industry fixed effects, and the inferences remain unaltered. The coefficient estimates range from 0.68 ($t = 1.71$) to 0.382

³⁰ In Table IA3, I follow Cheng et al. (2022) and adopt an alternative design. Specifically, I expand the analyses to all U.S. firms. As California firms are more likely to meet CCPA thresholds, I define firms headquartered in California as treated and those outside California as controls. With a larger sample size, the statistical power of the tests is higher; however, it may introduce noise because non-California firms may also be subject to the CCPA if they have significant operations in California. Nonetheless, the inferences are consistent under the alternative research design.

³¹ The correlations with analyst coverage and institutional ownership may be driven by the size effect, as firm size is highly correlated with analyst coverage (0.64) and institutional ownership (0.50) in a cross-sectional sample.

³² To better understand the determinants of cookie usage, in Table IA4 of the Internet Appendix, I regress the raw number of cookies on a series of firm characteristics. Firm size (*Size*), SG&A expenses (*SG&A*), and the indicator for being customer-oriented firms (*Customer*) are significant across different specifications, whereas most financial variables do not load. The findings imply that the use of cookies is primarily related to firms' data needs for their digital marketing and advertising because large firms, customer-oriented firms, and firms with more SG&A expenditures tend to have more demand for consumer data in their marketing and advertising activities. As the paper is based on a cross-sectional sample, the regression results based on the snapshot may not fully reveal the determinants of the use of cookies. Nonetheless, as the primary purpose of cookies is for digital marketing and advertising, it is unlikely that the usage of website cookies is determined by financial performance or firms' reporting purposes.

³³ Note that the sample size for Panel D is smaller than that for Panel C because not all firms in the sample are covered by the Aberdeen database.

³⁴ However, managing and analyzing collected data requires advanced data analytic technology, as shown in Table 7 and discussed in Section 6.4.

³⁵ As the dependent variable, *SF Freq*, is a count variable, I follow the suggestion from Cohn et al. (2022) and employ the Poisson regression model as a robustness check. The results are tabulated in Table IA5 in the Internet Appendix.

Table 2

Descriptive statistics.

This table tabulates the descriptive statistics of cookies and major variables used in the study. The sample is a cross-sectional sample based on website cookies collected in February 2023. Panel A shows the industry distribution of cookies. The industries are classified into consumer-oriented and non-consumer-oriented industries at the GICS four-digit industry level. For each industry, the table tabulates the average number of cookies and the average percentage of each category of cookies (classified based on ownership, duration, and purposes). Panel B tabulates the descriptive statistics of major variables used in the paper. Panels C and D present the correlation matrices. All variables are winsorized at the 1st and the 99th percentiles. * denotes statistical significance at the 5% level in Panels C and D. Variable definitions are available in [Appendix C](#).

Panel A: Industry Distribution of Cookies										
GICS Industries		# of Cookies	% First Party	% Third Party	% Session	% Persistent	% Necessary	% Functional	% Analytic	% Advertising
Consumer-Oriented Industries										
2550	Retailing	77.05	76.8%	23.2%	18.5%	81.5%	13.8%	5.8%	14.8%	45.5%
2030	Transportation	62.23	71.8%	28.2%	18.1%	81.9%	14.1%	4.8%	18.4%	42.4%
5010	Telecommunication Services	58.91	65.2%	34.8%	17.6%	82.4%	9.4%	2.9%	19.2%	53.0%
4510	Software and Services	50.49	79.6%	20.4%	16.9%	83.1%	18.0%	5.3%	17.7%	42.8%
4030	Insurance	47.39	76.9%	23.1%	18.3%	81.7%	22.5%	4.9%	21.1%	42.2%
2520	Consumer Durables and Apparel	46.38	79.2%	20.8%	22.1%	77.9%	19.9%	5.1%	18.7%	45.1%
4520	Technology Hardware and Equipment	43.43	75.2%	24.8%	18.2%	81.8%	17.3%	5.0%	18.9%	46.3%
2530	Consumer Services	40.18	74.9%	25.1%	19.5%	80.5%	18.9%	5.8%	15.0%	49.2%
5020	Media and Entertainment	37.58	77.8%	22.2%	18.9%	81.1%	19.4%	3.6%	17.4%	46.4%
2510	Automobiles and Components	33.17	78.6%	21.4%	19.4%	80.6%	12.7%	5.7%	16.2%	52.8%
3030	Household and Personal Products	29.29	84.6%	15.4%	21.7%	78.3%	31.6%	3.9%	19.9%	35.6%
3510	Healthcare Equipment and Service	27.46	80.1%	19.9%	17.9%	82.1%	12.8%	3.7%	19.8%	51.7%
3020	Food, Beverage, and Tobacco	26.79	81.2%	18.8%	20.4%	79.6%	13.2%	3.2%	20.0%	56.1%
3520	Pharmaceuticals	14.36	88.4%	11.6%	21.4%	78.6%	12.7%	2.5%	18.4%	54.8%
Consumer-Oriented Average		33.56	81.1%	18.9%	19.5%	80.5%	15.5%	4.0%	18.3%	49.6%
Non-Consumer-Oriented Industries										
4530	Semiconductor	43.60	80.6%	19.4%	19.0%	81.0%	20.7%	6.2%	18.7%	40.1%
5510	Utilities	41.66	78.5%	21.5%	21.7%	78.3%	16.4%	2.5%	19.8%	52.9%
2020	Commercial and Professional Services	41.07	75.3%	24.7%	17.5%	82.5%	22.5%	4.3%	14.0%	46.1%
4020	Diversified Financials	31.98	78.6%	21.4%	21.3%	78.7%	16.3%	5.3%	18.9%	47.2%
4010	Banks	30.36	68.2%	31.8%	21.8%	78.2%	13.8%	4.0%	13.7%	55.8%
2010	Capital Goods	28.85	80.1%	19.9%	20.3%	79.7%	17.6%	3.8%	19.9%	46.0%
1510	Materials	25.12	82.3%	17.7%	18.8%	81.2%	17.1%	3.1%	17.3%	53.2%
6020	Real Estate Management and Development	23.50	84.5%	15.5%	18.2%	81.8%	14.9%	3.5%	19.5%	45.3%
6010	Real Estate	18.50	82.3%	17.7%	25.9%	74.1%	10.9%	3.2%	23.1%	53.3%
1010	Energy	12.32	78.7%	21.3%	24.3%	75.7%	14.9%	2.3%	20.8%	54.8%
Non-Consumer-Oriented Average		29.25	77.4%	22.6%	21.0%	79.0%	16.6%	3.8%	18.0%	50.1%
Whole Sample Average		31.75	79.5%	20.5%	20.1%	79.9%	16.0%	3.9%	18.2%	49.8%

(continued on next page)

Table 2 (continued).

Panel B: Descriptive Statistics								
	N	Mean	SD	Min	25%	Median	75%	Max
<i>Voluntary Disclosure Proxies</i>								
SF Freq	2860	1.11	1.74	0.00	0.00	0.00	3.00	6.00
SF	2860	0.33	0.47	0.00	0.00	0.00	1.00	1.00
Customer Disc.	2119	1.88	2.34	0.00	0.25	1.05	2.63	15.03
Marketing Disc.	2119	1.06	1.11	0.00	0.11	0.75	1.61	6.46
Product Disc.	2119	1.61	2.12	0.00	0.17	0.93	2.38	43.36
Aggregated Disc.	2119	4.55	4.33	0.00	0.76	3.50	7.34	47.79
<i>Cookie Variables</i>								
Cookie	2860	2.88	1.02	1.10	2.20	2.77	3.58	5.48
Session Cookie	2860	1.33	0.84	0.00	0.69	1.10	1.95	3.76
Persistent Cookie	2860	2.67	1.05	1.10	1.95	2.56	3.37	5.36
FP Cookie	2860	2.58	1.00	0.00	1.95	2.48	3.22	5.25
TP Cookie	2860	1.31	1.27	0.00	0.00	1.10	2.20	4.54
Necessary Cookie	2860	1.09	1.07	0.00	0.00	0.69	1.79	4.78
Functional Cookie	2860	0.50	0.79	0.00	0.00	0.00	0.69	3.09
Analytic Cookie	2860	1.23	0.92	0.00	0.69	0.69	1.95	3.76
Advertising Cookie	2860	2.10	1.07	0.00	1.10	2.20	2.83	4.42
Unknown	2860	0.93	1.05	0.00	0.00	0.69	1.61	4.03
<i>Firm Characteristics</i>								
Size	2860	20.88	2.22	15.90	19.36	20.95	22.45	26.01
MB	2860	1.57	1.51	0.12	0.70	1.09	1.88	8.80
Leverage	2860	0.29	0.26	0.00	0.07	0.25	0.43	1.22
ROE	2860	−0.19	1.65	−9.91	−0.33	0.06	0.17	6.93
Loss	2860	0.45	0.50	0.00	0.00	0.00	1.00	1.00
Cash	2860	0.25	0.29	0.00	0.04	0.11	0.40	0.97
Capex	2860	0.03	0.04	0.00	0.00	0.02	0.04	0.23
PPE	2860	0.19	0.23	0.00	0.03	0.11	0.27	0.88
SG&A	2860	0.26	0.35	0.00	0.02	0.14	0.35	2.12
Return	2860	−0.21	0.44	−0.95	−0.56	−0.20	0.06	1.26
Volatility	2860	0.15	0.08	0.04	0.09	0.13	0.18	0.49
Customer	2860	0.58	0.49	0.00	0.00	1.00	1.00	1.00
HHI	2860	0.08	0.06	0.02	0.03	0.07	0.12	0.23
Analyst	2860	37.31	40.10	0.00	11.00	24.00	48.00	196.00
Inst	2860	0.65	0.31	0.00	0.42	0.74	0.91	1.00
Report Lag	2860	55.30	19.91	18.00	40.00	54.00	68.00	105.00
Restatement	2860	0.03	0.18	0.00	0.00	0.00	0.00	1.00
ICW	2860	0.26	0.83	0.00	0.00	0.00	0.00	5.00
<i>Aberdeen Data Analytic Variables</i>								
DBMS	1464	3.32	0.95	1.21	2.65	3.33	4.00	5.00
ERP	1464	4.16	0.83	2.00	3.66	4.33	5.00	5.00
SaaS	1437	3.11	0.86	1.92	2.44	3.00	3.67	5.00
Server	1860	0.91	2.94	0.00	0.01	0.06	0.39	22.27
Data Storage	1844	46.94	189.10	0.00	0.31	2.08	13.20	1526.65
IT Employee	1874	1.02	2.03	0.00	0.02	0.14	0.80	8.16
IT Budget	1752	3.79	17.38	0.00	0.05	0.50	1.91	155.56

(continued on next page)

Table 2 (continued).

Panel C: Pairwise Correlation for Major Variables																									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)
(1) Cookie	1.00																								
(2) SF Freq	0.20***	1.00																							
(3) SF	0.20***	0.91***	1.00																						
(4) Aggregated Disc.	0.30***	0.35***	0.34***	1.00																					
(5) Customer Disc.	0.35***	0.25***	0.24***	0.80***	1.00																				
(6) Marketing Disc.	0.24***	0.31***	0.30***	0.78***	0.54***	1.00																			
(7) Product Disc.	0.11***	0.28***	0.27***	0.75***	0.25***	0.48***	1.00																		
(8) Size	0.30***	0.19***	0.22***	−0.08***	−0.04	−0.06**	−0.08***	1.00																	
(9) MB	0.04	0.21***	0.20***	0.23***	0.18***	0.17***	0.18***	−0.08***	1.00																
(10) Leverage	0.10***	0.16***	0.17***	0.10***	0.10***	0.12***	0.03	0.14***	0.15***	1.00															
(11) ROE	0.07**	0.07**	0.07**	0.03	−0.00	0.05*	0.04	0.20***	0.05*	0.08***	1.00														
(12) Loss	−0.12***	−0.07**	−0.09***	0.06*	0.11***	0.02	−0.01	−0.52***	0.02	0.02	−0.25***	1.00													
(13) Cash	−0.19***	−0.14***	−0.17***	−0.04*	0.02	−0.09***	−0.07**	−0.51***	0.26**	−0.23***	−0.18***	0.51***	1.00												
(14) Capex	−0.02	−0.02	−0.01	−0.02	0.01	−0.00	0.04	0.03	0.14***	−0.00	−0.03	−0.14***	1.00												
(15) PPE	−0.05*	−0.08***	−0.08***	−0.10***	−0.10***	−0.06**	−0.06**	0.10***	−0.02	0.29**	0.04*	−0.10***	−0.28***	0.70***	1.00										
(16) SG&A	−0.07**	−0.05*	−0.08***	0.14***	0.15***	0.11***	0.08***	−0.62***	0.23**	0.02	−0.24***	0.46***	0.49***	−0.06**	−0.16***	1.00									
(17) Return	−0.04*	−0.01	0.01	−0.23***	−0.23***	−0.19***	−0.12***	0.39***	0.11***	−0.06**	0.19***	−0.50***	−0.31***	0.07**	0.19***	−0.43***	1.00								
(18) Volatility	−0.17***	−0.12***	−0.15***	−0.03	0.02	−0.02	−0.06**	−0.46***	0.08***	0.06**	−0.16***	0.49***	0.42***	0.10***	0.03	0.41***	−0.25***	1.00							
(19) Customer	0.06**	0.13***	0.10***	0.24***	0.21***	0.25***	0.13***	−0.26***	0.21***	0.15***	−0.12***	0.35***	0.37***	−0.06**	−0.09***	0.39***	−0.34***	0.23***	1.00						
(20) HHI	0.11***	−0.06*	−0.09***	0.07**	0.11***	0.03	0.00	0.07**	−0.09***	−0.16***	0.03	−0.06**	−0.06**	−0.16***	−0.23***	−0.01	−0.01	−0.16***	0.20***	1.00					
(21) Analyst	0.24***	0.17***	0.19***	0.04	0.06**	0.03	0.00	0.67***	0.19***	0.18***	0.09***	−0.23***	−0.16***	0.14***	0.17***	−0.21***	0.19***	−0.18***	−0.03	−0.09***	1.00				
(22) Inst	0.18***	0.37***	0.39***	0.09***	0.10***	0.07**	0.04	0.50***	0.16***	0.17***	0.10***	−0.21***	−0.15***	0.06**	0.10***	−0.28***	0.22***	−0.19***	−0.02	−0.14***	0.39***	1.00			
(23) Report Lag	−0.29***	−0.18***	−0.21***	0.03	0.00	0.06**	0.02	−0.75***	−0.01	0.05*	−0.21***	0.53***	0.40***	0.08***	0.06**	0.48***	−0.39***	0.52***	0.31***	−0.23***	−0.43***	−0.40***	1.00		
(24) Restatement	0.02	0.00	−0.01	0.02	0.02	0.02	−0.01	−0.04	−0.04*	0.02	−0.04	0.04	−0.03	0.02	−0.01	0.03	−0.07**	0.04*	0.05*	−0.04	−0.04	−0.07**	0.08***	1.00	
(25) ICW	−0.00	−0.03	−0.05*	0.05*	0.06**	0.06**	0.01	−0.17***	−0.04*	0.02	−0.09***	0.16***	0.04	0.04*	0.01	0.12***	−0.19***	0.15***	0.06**	−0.04	−0.13***	−0.11***	0.28***	0.25***	1.00

Panel D: Pairwise Correlation between Disclosure Proxies and Data Analytics Variables														
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
(1) Cookie	1.00													
(2) SF Freq	0.18***	1.00												
(3) SF	0.18***	0.89***	1.00											
(4) Aggregated Disc.	0.22***	0.43***	0.41***	1.00										
(5) Customer Disc.	0.28***	0.32***	0.31***	0.79***	1.00									
(6) Marketing Disc.	0.15***	0.33***	0.31***	0.78***	0.51***	1.00								
(7) Product Disc.	0.07*	0.36***	0.33***	0.78***	0.30***	0.51***	1.00							
(8) DBMS	0.07*	0.18***	0.16***	0.18***	0.22***	0.09**	0.10**	1.00						
(9) ERP	−0.08**	0.17***	0.16***	0.28***	0.11***	0.22***	0.33***	0.26***	1.00					
(10) SaaS	0.07*	0.12***	0.10**	0.16***	0.24***	0.06*	0.05	0.49***	0.08**	1.00				
(11) Server	−0.05	−0.08**	−0.08*	0.12***	0.08**	0.07*	0.11***	0.07*	0.05	0.08**	1.00			
(12) Data Storage	−0.07*	−0.10***	−0.09**	0.05	0.06	0.04	0.01	0.07*	0.05	0.04	0.64***	1.00		
(13) IT Employee	−0.07*	−0.07*	−0.06*	0.05	−0.02	0.03	0.10**	0.10**	0.02	0.14***	0.38***	0.17***	1.00	
(14) IT Budget	0.04	0.03	0.02	0.02	0.04	−0.02	0.02	0.07*	0.00	0.04	0.01	0.03	0.04	1.00

Table 3

Baseline results: Cookies and voluntary disclosure.

This table displays the results of the baseline regressions. The dependent variable is the frequency of management sales forecasts (*SF Freq*), the indicator of issuing management sales forecasts (*SF*), and FinBERT-based textual analysis measures for Panels A, B, and C, respectively. For each panel, I tabulate the results under specifications with different sets of control variables, as well as with and without industry fixed effects (GICS industry group level). When including data analytic technology variables, I include these variables with missing values imputed as zero as well as indicators for variables being missing. The variables are as defined in [Appendix C](#). All variables are winsorized at the 1st and the 99th percentiles. *t*-statistics are reported in parentheses and are based on standard errors clustered at the industry level. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

Panel A: The Frequency of Management Sales Forecasts						
	SF Freq					
	(1)	(2)	(3)	(4)	(5)	(6)
Cookie	0.382*** (4.69)	0.122** (2.29)	0.148*** (3.15)	0.126** (2.56)	0.068* (1.71)	0.073* (1.89)
Size		−0.089** (−2.05)	−0.065* (−1.69)	−0.060 (−1.44)	−0.023 (−0.64)	−0.000 (−0.00)
MB		0.195*** (5.31)	0.192*** (4.54)	0.175*** (4.78)	0.072** (2.54)	0.068** (2.40)
Leverage		0.065 (0.32)	0.223 (1.03)	0.124 (0.61)	0.322** (2.34)	0.319** (2.16)
ROE		0.020 (1.06)	0.019 (0.98)	0.019 (1.02)	−0.005 (−0.30)	−0.004 (−0.24)
Loss		0.052 (0.49)	0.067 (0.69)	0.023 (0.23)	0.069 (0.80)	0.052 (0.61)
Cash		−1.786*** (−7.65)	−1.314*** (−6.57)	−1.544*** (−6.53)	−1.363*** (−7.51)	−1.210*** (−6.63)
Capex		2.953*** (2.72)	2.882*** (2.79)	3.134*** (3.06)	2.027*** (2.71)	2.153*** (2.97)
PPE		−1.492*** (−5.70)	−1.495*** (−5.53)	−1.538*** (−6.11)	−1.024*** (−4.21)	−1.005*** (−4.40)
SG&A		0.241** (2.02)	0.304*** (2.92)	0.249** (2.35)	−0.005 (−0.04)	0.021 (0.19)
Return		−0.321*** (−3.15)	−0.430*** (−4.36)	−0.350*** (−3.41)	−0.079 (−0.82)	−0.094 (−0.97)
Volatility		−0.593 (−0.97)	−0.403 (−0.76)	−0.399 (−0.73)	−0.738 (−1.30)	−0.608 (−1.16)
Report Lag		−0.010** (−2.29)	−0.005 (−0.77)	−0.008** (−2.17)	−0.013*** (−4.12)	−0.012*** (−3.77)
Restatement		0.100 (0.70)	0.144 (1.03)	0.086 (0.62)	0.142 (1.08)	0.128 (0.99)
ICW		0.035 (0.72)	0.033 (0.68)	0.047 (1.00)	0.022 (0.47)	0.029 (0.63)
Analyst		0.002 (0.88)	0.002 (0.92)	0.001 (0.63)	0.001 (1.00)	0.001 (0.74)
Inst		1.751*** (6.12)	1.682*** (6.39)	1.560*** (6.08)	1.254*** (4.76)	1.135*** (4.57)
Customer		0.581** (2.21)		0.522** (2.13)		
HHI		−1.669 (−0.78)		−1.645 (−0.85)		
Observations	2860	2860	2860	2860	2860	2860
Adjusted R^2	0.050	0.243	0.257	0.270	0.381	0.391
Industry FE	No	No	No	No	Yes	Yes
Data Analytic Variables	No	No	Yes	Yes	No	Yes

(continued on next page)

($t = 4.69$), suggesting that a 10% increase in cookies (roughly three cookies), *ceteris paribus*, is associated with a 0.0065 to 0.0364 increase in forecast frequency, roughly 0.6% to 3.2% of the mean. The effect is economically meaningful.³⁶

Although I have controlled for a comprehensive list of variables and included industry fixed effects, it is still conceivable that the results may be affected by correlated omitted variables. To minimize the concern, I follow [Larcker and Rusticus \(2010\)](#) and evaluate the extent of the impact of potential omitted-variable bias using the impact threshold of confounding variables (ITCV)

³⁶ To make the magnitude more sensible, I run regressions with standardized independent variables (with mean of zero and standard deviation of one) in Table IA6 of the Internet Appendix. The effect of *Cookie* on the frequency of sales forecasts is relatively large compared with other firm characteristics, suggesting that the effects of cookies on disclosure are not trivial.

Table 3 (continued).

Panel B: The Incidence of Management Sales Forecasts						
	SF					
	(1)	(2)	(3)	(4)	(5)	(6)
Cookie	0.033** (2.09)	0.034** (2.37)	0.016** (2.01)	0.185** (2.08)	0.202** (2.29)	0.133* (1.67)
Size	−0.027** (−2.16)	−0.018 (−1.58)	−0.001 (−0.07)	−0.143* (−1.95)	−0.103 (−1.35)	−0.009 (−0.10)
MB	0.048*** (4.74)	0.043*** (4.32)	0.013** (2.34)	0.272*** (5.09)	0.248*** (4.39)	0.038 (0.76)
Leverage	0.042 (0.74)	0.059 (1.06)	0.102*** (3.13)	0.312 (0.91)	0.399 (1.19)	0.915*** (3.04)
ROE	0.005 (1.22)	0.005 (1.11)	−0.001 (−0.31)	0.034 (1.06)	0.025 (0.78)	−0.001 (−0.02)
Loss	0.011 (0.44)	0.007 (0.29)	0.022 (1.09)	0.168 (1.40)	0.144 (1.22)	0.311** (2.42)
Cash	−0.504*** (−7.87)	−0.435*** (−6.63)	−0.329*** (−7.66)	−3.348*** (−5.49)	−3.142*** (−4.97)	−2.399*** (−4.49)
Capex	0.759*** (2.68)	0.811*** (2.98)	0.558** (2.41)	5.740*** (2.82)	5.772*** (2.92)	4.122** (2.23)
PPE	−0.428*** (−6.13)	−0.442*** (−6.78)	−0.315*** (−5.32)	−2.805*** (−6.16)	−2.926*** (−6.45)	−2.058*** (−5.08)
SG&A	0.066* (1.96)	0.067** (2.22)	0.008 (0.29)	0.600** (2.28)	0.680*** (2.61)	0.228 (0.87)
Return	−0.076** (−2.55)	−0.087*** (−2.93)	−0.015 (−0.66)	−0.423** (−2.08)	−0.455** (−2.25)	0.043 (0.22)
Volatility	−0.240 (−1.32)	−0.184 (−1.12)	−0.271** (−2.44)	−1.189 (−0.75)	−0.759 (−0.48)	−1.285 (−0.77)
Report Lag	−0.003** (−2.27)	−0.002** (−2.14)	−0.004*** (−5.27)	−0.021*** (−2.72)	−0.019** (−2.43)	−0.028*** (−3.88)
Restatement	0.001 (0.03)	−0.002 (−0.05)	0.007 (0.19)	0.043 (0.18)	−0.037 (−0.15)	0.083 (0.28)
ICW	0.005 (0.42)	0.009 (0.74)	0.003 (0.38)	0.047 (0.56)	0.080 (0.91)	0.065 (0.70)
Analyst	0.001 (1.15)	0.000 (0.94)	0.000 (1.31)	0.002 (0.84)	0.002 (0.56)	0.003 (1.30)
Inst	0.487*** (6.70)	0.436*** (6.64)	0.311*** (10.73)	3.312*** (13.91)	3.056*** (12.53)	2.829*** (10.65)
Customer	0.155** (2.07)	0.138* (1.96)		0.867* (1.96)	0.821* (1.91)	
HHI	−0.687 (−1.10)	−0.668 (−1.16)		−4.157 (−1.07)	−4.096 (−1.10)	
Observations	2860	2860	2860	2860	2860	2860
Adjusted R^2	0.260	0.286	0.416	0.250	0.281	0.439
Industry FE	No	No	Yes	No	No	Yes
Model	OLS	OLS	OLS	Logistic	Logistic	Logistic
Data Analytic Variables	No	Yes	Yes	No	Yes	Yes

(continued on next page)

approach.³⁷ I calculate the ITCV based on column (4), which includes firm-level controls but no fixed effects. The ITCV for *Cookie* is 0.0387, meaning that an omitted variable would need a joint partial-correlation product with *SF Freq* and *Cookie* exceeding 0.0387 to invalidate the inference. To gauge whether this threshold is large enough to rule out the omitted variable concern, I compare it with the products of partial correlations for existing control variables. Among all existing control variables, *Size* and *MB* have the largest products of partial correlations of 0.0162 and 0.0151, respectively. Thus, an omitted variable would have to be more strongly correlated with both disclosure frequency and cookie usage than these two key firm characteristics. As it is unlikely that an omitted variable is as important as firm size and market-to-book ratios in explaining the variation in the dependent variable and *Cookie*, omitted variable bias is unlikely to overturn the results.³⁸

³⁷ The ITCV is the smallest product of partial correlations that an omitted variable must have with dependent and independent variables in the test to make a significant coefficient on the independent variable insignificant. For example, if the ITCV value of the test is 0.01, an omitted confounding variable's partial correlation with the dependent variable and with the independent variable must have a product greater than 0.01 to affect the inferences.

³⁸ I also conduct the analyses based on ITCV for regressions in Panels B and C and find no indication that omitted variables may affect my inferences.

Table 3 (continued).

Panel C: Discussions in 10-K Filings Regarding Customers, Marketing, and Products								
	Customer Disc.		Marketing Disc.		Product Disc.		Aggregated Disc.	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Cookie	0.674*** (6.36)	0.336*** (7.13)	0.198*** (4.96)	0.146*** (6.00)	0.162** (2.43)	0.109** (2.61)	1.034*** (6.06)	0.591*** (6.78)
Size	-0.235*** (-3.16)	-0.147*** (-2.87)	-0.144*** (-4.46)	-0.137*** (-5.27)	-0.305*** (-2.95)	-0.218** (-2.64)	-0.684*** (-4.25)	-0.501*** (-4.22)
MB	0.218** (2.05)	0.084* (1.89)	0.108*** (5.22)	0.051*** (2.75)	0.230*** (5.58)	0.079* (1.92)	0.556*** (4.35)	0.214*** (2.69)
Leverage	0.365 (1.43)	-0.032 (-0.16)	-0.010 (-0.07)	-0.087 (-0.98)	-0.382 (-1.53)	-0.233 (-1.49)	-0.027 (-0.06)	-0.352 (-1.10)
ROE	0.018 (0.46)	0.007 (0.29)	0.038** (2.60)	0.017 (1.35)	0.046* (1.71)	0.004 (0.17)	0.102 (1.61)	0.028 (0.61)
Loss	0.124 (0.39)	0.090 (0.56)	-0.143* (-1.70)	-0.080 (-1.20)	-0.150 (-0.94)	-0.144 (-1.05)	-0.169 (-0.46)	-0.135 (-0.58)
Cash	-1.196* (-1.96)	-0.005 (-0.01)	-1.420*** (-4.76)	-0.700*** (-4.53)	-2.061*** (-4.56)	-1.476*** (-4.14)	-4.678*** (-4.04)	-2.181*** (-3.06)
Capex	3.739** (2.34)	3.270*** (3.14)	2.355*** (3.01)	1.128* (1.72)	2.829* (1.86)	0.283 (0.23)	8.923*** (2.97)	4.681** (2.25)
PPE	-1.349*** (-3.07)	-1.442*** (-4.45)	-1.028*** (-4.51)	-0.959*** (-4.35)	-1.445*** (-3.70)	-0.903** (-2.51)	-3.822*** (-4.59)	-3.304*** (-4.79)
SG&A	0.075 (0.18)	0.237 (1.00)	-0.099 (-0.58)	-0.057 (-0.40)	-0.208 (-0.80)	-0.168 (-0.95)	-0.232 (-0.32)	0.012 (0.03)
Return	-0.931*** (-4.99)	-0.552*** (-3.73)	-0.449*** (-6.66)	-0.265*** (-3.79)	-0.770*** (-4.33)	-0.377** (-2.57)	-2.150*** (-6.58)	-1.194*** (-4.29)
Volatility	-0.181 (-0.12)	0.174 (0.12)	-0.364 (-0.84)	-0.026 (-0.08)	-1.743*** (-2.67)	-2.032*** (-3.71)	-2.288 (-1.10)	-1.885 (-1.06)
Report Lag	-0.002 (-0.34)	-0.011** (-2.37)	0.003 (0.80)	-0.002 (-1.28)	-0.003 (-0.52)	-0.003 (-0.65)	-0.003 (-0.22)	-0.017* (-1.95)
Restatement	0.039 (0.16)	0.047 (0.20)	-0.083 (-0.62)	-0.073 (-0.68)	-0.259 (-1.16)	-0.217 (-1.21)	-0.304 (-0.63)	-0.242 (-0.63)
ICW	0.070 (1.18)	0.091 (1.54)	0.032 (0.99)	0.026 (0.90)	0.015 (0.30)	0.012 (0.24)	0.116 (1.04)	0.129 (1.24)
Analyst	0.004** (2.11)	0.003 (1.66)	0.003** (2.56)	0.002*** (2.97)	0.005* (1.86)	0.004** (2.04)	0.012*** (2.71)	0.009*** (2.96)
Inst	1.099*** (3.27)	0.495** (2.63)	0.392*** (4.45)	0.236** (2.35)	0.472** (2.20)	0.097 (0.58)	1.963*** (4.44)	0.828** (2.63)
Customer	0.281 (0.87)		0.514*** (2.88)		0.484 (1.38)		1.278* (1.79)	
HHI	3.552 (1.09)		-0.252 (-0.17)		-0.456 (-0.17)		2.844 (0.45)	
Observations	2119	2119	2119	2119	2119	2119	2119	2119
Adjusted R^2	0.252	0.536	0.248	0.458	0.171	0.368	0.309	0.548
Industry FE	No	Yes	No	Yes	No	Yes	No	Yes
Data Analytic Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

In Panel B of Table 3, I replace the dependent variable with an indicator of issuing management sales forecasts and estimate OLS regressions in columns (1) to (3) and logistic regressions in columns (4) to (6). Similar to Panel A, I find that the number of cookies (*Cookie*) is positively associated with the likelihood of issuing management sales forecasts. The inferences are robust to different sets of control variables, both the linear probability model and the logistic model, and the inclusion of industry fixed effects.

Panel C reports the results using FinBERT-based qualitative disclosure proxies as dependent variables.³⁹ In columns (1) and (2), I regress the proxy for customer-related disclosure (*Customer Disc*) on *Cookie*. Column (1) controls for firm and industry characteristics, while column (2) further includes industry fixed effects. The coefficient on *Cookie* (0.336, $t = 7.13$) implies that a 10% increase in cookies is associated with a 0.032 increase in customer-related sentences (about a 1.8% rise relative to the sample mean).

Columns (3) to (6) replace the dependent variable with marketing-related and product-related disclosure measures (*Marketing Disc* and *Product Disc*) and show positive but weaker associations. This finding suggests that cookies capture more customer-related information than marketing- or product-specific information. Columns (7) and (8) combine all three categories into an aggregate disclosure measure (*Aggregated Disc*), the sum of the percentage of 10-K sentences discussing customers, marketing, or products, to

³⁹ To ensure robustness to the FinBERT model, I replicate the analyses using a simple bag-of-words approach and report the dictionary and regression results in Table IA7 of the Internet Appendix. The inferences remain unchanged.

Table 4

First- vs. Third-party cookies.

This table reports the results of the regressions of voluntary disclosure proxies on the numbers of first- and third-party cookies (*FP Cookie* and *TP Cookie*). The dependent variables are the frequency of management sales forecasts (*SF Freq*), the indicator for issuing sales forecasts (*SF*), FinBERT-based measure for customer-related disclosure (*Customer Disc.*), and aggregated FinBERT-based measure for qualitative disclosure about customers, marketing, and products (*Aggregate Disc.*). Control variables are included for all columns, and they are as defined in [Appendix C](#). When including data analytic technology variables, I include these variables with missing values imputed as zero as well as indicators for variables being missing. All variables are winsorized at the 1st and the 99th percentiles. *t*-statistics are reported in parentheses and are based on standard errors clustered at the industry level. *F*-statistics testing whether the coefficients of *FP Cookie* and *TP Cookie* are equal are tabulated at the bottom of the table. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

	SF Freq	SF	Customer Disc.	Aggregated Disc.
	(1)	(2)	(3)	(4)
FP Cookie	0.119*** (3.24)	0.029*** (2.92)	0.562*** (5.22)	0.926*** (6.11)
TP Cookie	0.028 (0.93)	0.010 (1.14)	0.262*** (4.45)	0.322*** (2.91)
Observations	2471	2471	1879	1879
Adjusted <i>R</i> ²	0.266	0.280	0.262	0.310
Firm Characteristics Variables	Yes	Yes	Yes	Yes
Data Analytic Variables	Yes	Yes	Yes	Yes
F-Statistics (FP Cookie = TP Cookie)	4.74**	2.19	8.61***	13.90***

account for sentence overlap. The inferences remain consistent. In the analyses below, I only tabulate the results based on *Customer Disc* and *Aggregated Disc* for brevity.

These findings indicate that managers place higher weights on customer-, marketing-, and product-related topics in 10-K filings when cookies improve the internal information quality of these topics. In addition, Panel C rules out the possibility that the number of cookies simply proxies for the volume of any other information collected by firms.⁴⁰

To summarize, the baseline analyses indicate that cookie-collected data are useful for voluntary disclosure. With more granular and timely information about customers, firms provide more management sales forecasts as well as larger percentages of qualitative discussions on customers, marketing, and products in 10-K filings.

6.3. Cookie characteristics and voluntary disclosure

Building upon the baseline results, I further exploit the rich variation in the ownership, the purpose, and the duration of cookies to examine whether the relevance and the volume of data collected through cookies relate to the usefulness of cookies in voluntary disclosure.

6.3.1. The relevance of data collected by cookies

I exploit the ownership of cookies to examine the relevance of cookie-collected data. As discussed in [Section 4.1](#), I expect that, relative to third-party cookies, first-party cookies are directly managed by firms and tend to collect data that are more relevant and specific to firms' data needs. Thus, I expect that the relation between cookies and disclosure is more pronounced when firms use more first-party cookies.

In [Table 4](#), I conduct a horse race between first- and third-party cookies (*FP Cookie* and *TP Cookie*). I find that only *FP Cookie* consistently shows positive and significant relations with all three voluntary disclosure proxies under specifications with and without control variables. In contrast, *TP Cookie* exhibits significantly weaker results, with coefficients significant only in columns (3) and (4) when using proxies for qualitative disclosures as the dependent variable. The results indicate that third-party cookies are still informative but to a lesser extent compared with first-party cookies. *F*-statistics suggest that the two coefficients are significantly different.

Among all types of cookies, analytic cookies are specifically designed to capture consumer preferences and behavioral patterns. Such data are useful in forecasting future sales and provide granular inputs for managers' qualitative disclosures about customers, marketing, and products. Hence, I expect that analytic cookies provide more relevant information for financial reporting than other types of cookies.

[Table 5](#) examines cookies with different purposes based on the GDPR classification. In columns (1), (3), (5), and (7), *Analytic Cookie* is positively and significantly associated with disclosure proxies. In the specifications including all types of cookies, *Analytic*

⁴⁰ If cookies are a proxy for the volume of any information collected by firms, one would not expect increases in the *percentages* of customer-, marketing-, and product-related discussions, but instead, an overall increase in the volume of information disclosed in 10-K filings. In untabulated analyses, I further rule out this possibility by showing that the number of cookies is *not* associated with the length of 10-K filings.

Table 5

Cookies with different purposes.

This table presents the results of the regression of voluntary disclosure proxies on the numbers of cookies with different purposes (*Analytic Cookie*, *Necessary Cookie*, *Functional Cookie*, *Advertising Cookie*, and *Unknown*). The dependent variables are the frequency of management sales forecasts (*SF Freq*), the indicator for issuing sales forecasts (*SF*), FinBERT-based measure for customer-related disclosure (*Customer Disc.*), and aggregated FinBERT-based measure for qualitative disclosure about customers, marketing, and products (*Aggregate Disc.*). Control variables are included for all columns, and they are as defined in [Appendix C](#). When including data analytic technology variables, I include these variables with missing values imputed as zero as well as indicators for variables being missing. All variables are winsorized at the 1st and the 99th percentiles. *t*-statistics are reported in parentheses and are based on standard errors clustered at the industry level. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

	SF Freq		SF		Customer Disc.		Aggregated Disc.	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Analytic Cookie	0.097** (2.59)	0.072* (1.69)	0.022** (2.11)	0.012 (1.03)	0.549*** (4.92)	0.216** (2.27)	0.887*** (5.02)	0.429*** (2.72)
Functional Cookie		0.072 (1.52)		0.013 (0.96)		0.225*** (3.48)		0.345*** (2.91)
Advertising Cookie		−0.027 (−0.53)		−0.002 (−0.14)		0.186*** (3.63)		0.204* (1.88)
Necessary Cookie		0.034 (0.92)		0.013 (1.19)		0.200*** (3.53)		0.362*** (4.07)
Unknown		0.011 (0.26)		0.007 (0.66)		0.245*** (4.14)		0.311*** (3.15)
Observations	2471	2471	2471	2471	1879	1879	1879	1879
Adjusted <i>R</i> ²	0.265	0.265	0.277	0.278	0.227	0.270	0.290	0.315
Firm Characteristics Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Data Analytic Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F-Statistics (Analytic = (Functional + Advertising + Necessary)/3)		0.60		0.83		0.02		0.53

Cookie remains significant in most cases (except for column (4)), although F-tests indicate that the effect is not statistically larger than the average effects of other cookie types. The insignificant differences between analytic cookies and other cookies likely reflect that other types of cookies also provide relevant information for disclosures. For example, advertising and functional cookies also capture detailed information about browsing histories and repeat visits, which can be informative about consumer demand and engagement.

6.3.2. The volume of data collected by cookies

The usefulness of cookie-collected data also depends on their volume. I make use of the duration of cookies to *indirectly* capture the volume of data collected. As persistent cookies last for a longer period on users' devices than session cookies and allow firms to collect time-series data about the same customer, I expect that persistent cookies provide a larger volume of data than session cookies.

In [Table 6](#), I show that persistent cookies exhibit a positive and significant association with disclosure proxies in all specifications. Although session cookies are significantly associated with qualitative disclosure proxies, the F-tests suggest that the effect of persistent cookies is significantly stronger. The results indicate that persistent cookies collect a larger volume of data, thus having stronger effects on internal information quality and voluntary disclosures.

In sum, these findings suggest that cookies tend to be more useful for disclosure if they collect data that are more relevant to firms' financial reporting or if they collect data of a larger volume. The findings also further validate the baseline findings in [Table 3](#).

6.4. The role of data analytic technology

The usefulness of cookie-collected data depends not only on the data *per se* but also on firms' ability to manage and analyze them ([LaValle et al., 2010](#); [Kiron et al., 2013](#)). Firms with stronger data analytic technology are likely to extract greater informational value from cookie-collected data. To test this idea, I consider three different dimensions of data analytic technology, including (1) hardware capacity, (2) software availability, and (3) general data-analytic capability (IT human capital and investment).

Panel A of [Table 7](#) reports results for hardware capacity, measured by data storage (*Data Storage*) and the number of servers (*Server*). The relation between cookies and disclosure is positive but weak among firms with low hardware capacity. In contrast, the interaction terms show significantly stronger effects for firms with high hardware capacity, indicating that computing resources enhance firms' ability to use cookie data effectively.⁴¹

⁴¹ The number of observations varies across columns due to missing values in the Aberdeen database variables.

Table 6

Cookies of different lengths of duration.

This table shows the results of the regression of voluntary disclosure proxies on the numbers of session and persistent cookies (*Session Cookie* and *Persistent Cookie*). The dependent variables are the frequency of management sales forecasts (*SF Freq*), the indicator for issuing sales forecasts (*SF*), FinBERT-based measure for customer-related disclosure (*Customer Disc.*), and aggregated FinBERT-based measure for qualitative disclosure about customers, marketing, and products (*Aggregate Disc.*). Control variables are included for all columns, and they are as defined in [Appendix C](#). When including data analytic technology variables, I include these variables with missing values imputed as zero as well as indicators for variables being missing. All variables are winsorized at the 1st and the 99th percentiles. *t*-statistics are reported in parentheses and are based on standard errors clustered at the industry level. *F*-statistics testing whether the coefficients of *Persistent Cookie* and *Session Cookie* are equal are tabulated at the bottom of the table. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

	SF Freq	SF	Customer Disc.	Aggregated Disc.
	(1)	(2)	(3)	(4)
Persistent Cookie	0.123** (2.09)	0.045*** (3.02)	0.619*** (5.71)	0.907*** (5.28)
Session Cookie	0.016 (0.31)	−0.012 (−0.97)	0.272*** (4.06)	0.442*** (3.04)
Observations	2471	2471	1879	1879
Adjusted <i>R</i> ²	0.266	0.281	0.263	0.310
Firm Characteristics Variables	Yes	Yes	Yes	Yes
Data Analytic Variables	Yes	Yes	Yes	Yes
F-Statistics (Persistent = Session)	1.30	6.67**	9.63 ***	4.36 **

Table 7

Data analytic technology and the effects of cookie-collected data.

This table tabulates the regression results regarding the effect of data analytic technology on the usefulness of cookies in voluntary disclosure. Panel A evaluates the hardware capability of data analytic technology, including the capacity of data storage and the number of servers (*Data Storage* and *Server*). Panel B tabulates the results for the software aspect of data analytic technology, including the rating for DBMS, ERP, and SaaS (*DBMS*, *ERP* and *SaaS*). Panel C focuses on the percentage IT employees (*IT Employee*) and the percentage of IT budget as of sales revenue (*IT Budget*). For each panel, the dependent variables are the frequency of management sales forecasts (*SF Freq*), the indicator for issuing sales forecasts (*SF*), FinBERT-based measure for customer-related disclosure (*Customer Disc.*), and aggregated FinBERT-based measure for qualitative disclosure about customers, marketing, and products (*Aggregate Disc.*). The variables of interest are the interaction terms between the number of cookies (*Cookie*) and the indicator that data analytic technology proxies are larger than the median. Control variables are included for all columns, and they are as defined in [Appendix C](#). When including data analytic technology variables, I include these variables with missing values imputed as zero as well as indicators for variables being missing. All variables are winsorized at the 1st and the 99th percentiles. *t*-statistics are reported in parentheses and are based on standard errors clustered at the industry level. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

Panel A: The Role of Hardware Capability

	SF Freq		SF		Customer Disc.		Aggregated Disc.	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(Data Storage ≥ Median) × Cookie	0.220** (2.51)		0.064*** (2.82)		0.750*** (4.74)		1.228*** (5.57)	
(Servers ≥ Median) × Cookie		0.288*** (3.83)		0.078*** (3.84)		0.665*** (4.77)		1.153*** (4.45)
Cookie	0.071 (0.89)	0.045 (0.67)	0.019 (0.94)	0.013 (0.74)	0.258*** (2.88)	0.281*** (3.11)	0.384** (2.41)	0.386** (2.26)
Data Storage ≥ Median	−0.820*** (−3.63)		−0.217*** (−3.60)		−1.704*** (−4.82)		−3.830*** (−5.74)	
Servers ≥ Median		−0.712*** (−3.43)		−0.212*** (−3.73)		−1.604*** (−4.39)		−3.307*** (−4.47)
Observations	1844	1860	1844	1860	1393	1403	1393	1403
Adjusted <i>R</i> ²	0.260	0.265	0.265	0.268	0.270	0.265	0.304	0.305
Firm Characteristics Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

(continued on next page)

In Panel B, the interaction terms are positive and significant for *DBMS* and *SaaS*, suggesting that cookie-collected data are more useful when firms have better software to manage and analyze them. The effect is weaker for ERP systems potentially because they are primarily for information sharing rather than analytics.

Panel C examines general data analytic capabilities, the share of IT employees (*IT Employee*) and IT budget (*IT Budget*). Firms with greater IT human capital and investment make more effective use of cookie-collected data in disclosure, though the effect for *IT Budget* is less robust.

Table 7 (continued).

Panel B: The Role of Software Availability												
	SF Freq			SF			Customer Disc.			Aggregated Disc.		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
(DBMS $\geq$ Median) $\times$ Cookie	0.196** (2.08)			0.042 (1.64)			0.349*** (3.15)			0.531** (2.53)		
(ERP $\geq$ Median) $\times$ Cookie		0.053 (0.55)			0.018 (0.72)			−0.052 (−0.42)			0.070 (0.28)	
(SaaS $\geq$ Median) $\times$ Cookie			0.110 (1.10)			0.020 (0.76)			0.489*** (2.91)			0.673** (2.22)
Cookie	0.028 (0.41)	0.111 (1.56)	0.069 (0.89)	0.017 (0.79)	0.032 (1.48)	0.027 (1.23)	0.280*** (3.67)	0.469*** (4.35)	0.228*** (3.22)	0.431*** (2.77)	0.667*** (3.65)	0.376* (1.84)
DBMS $\geq$ Median	−0.344 (−1.23)			−0.082 (−1.08)			−0.583* (−1.84)			−0.954 (−1.40)		
ERP $\geq$ Median		0.280 (1.18)			0.061 (0.87)			0.279 (0.80)			1.278* (1.80)	
SaaS $\geq$ Median			−0.175 (−0.63)			−0.029 (−0.38)			−1.021** (−2.15)			−1.760* (−1.90)
Observations	1464	1464	1437	1464	1464	1437	1152	1152	1130	1152	1152	1130
Adjusted R^2	0.262	0.268	0.261	0.254	0.262	0.250	0.242	0.225	0.250	0.311	0.330	0.311
Firm Characteristics Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Panel C: The Role of IT Employees and Budget												
	SF Freq		SF		Customer Disc.		Aggregated Disc.					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)				
(IT Employee $\geq$ Median) $\times$ Cookie	0.119 (1.54)		0.042** (2.05)		0.345** (2.42)		0.493** (2.17)					
(IT Budget $\geq$ Median) $\times$ Cookie		0.043 (0.54)		0.031 (1.41)		0.302** (2.09)		0.402** (2.03)				
Cookie	0.132* (1.99)	0.168** (2.40)	0.032* (1.78)	0.036* (1.88)	0.420*** (4.61)	0.438*** (4.84)	0.676*** (3.90)	0.655*** (3.80)				
IT Employee $\geq$ Median	−0.450** (−2.19)		−0.152*** (−2.83)		−0.912** (−2.49)		−1.738*** (−2.87)					
IT Budget $\geq$ Median		−0.150 (−0.63)		−0.110* (−1.74)		−1.032*** (−2.93)		−2.039*** (−3.54)				
Observations	1874	1752	1874	1752	1392	1309	1392	1309				
Adjusted R^2	0.261	0.266	0.265	0.270	0.250	0.245	0.296	0.300				
Firm Characteristics Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes				

Table 8

Industry competition and the effects of cookie-collected data.

This table reports the results examining the role of industry competition. The dependent variable is the frequency of management sales forecasts (*SF Freq*). Columns (1) and (2) tabulates the results for five industries with the lowest HHI, and columns (3) and (4) shows the results for all other industries. Control variables are included for all columns, and they are as defined in [Appendix C](#). When including data analytic technology variables, I include these variables with missing values imputed as zero as well as indicators for variables being missing. All variables are winsorized at the 1st and the 99th percentiles. *t*-statistics are reported in parentheses and are based on standard errors clustered at the industry level. *F*-statistics testing whether the coefficients of *FP Cookie* and *TP Cookie* are equal are tabulated at the bottom of the table. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

	Competitive Industries		Other Industries	
	(1)	(2)	(3)	(4)
Cookie	0.080 (1.01)	0.027 (0.40)	0.180*** (3.19)	0.095** (2.04)
Observations	869	869	1991	1991
Adjusted R^2	0.182	0.258	0.317	0.447
Firm Characteristics Variables	Yes	Yes	Yes	Yes
Data Analytic Variables	Yes	Yes	Yes	Yes
Industry FE	No	Yes	No	Yes

Overall, these results indicate that data analytic technology strengthens firms' ability to convert cookie-collected data into internal information useful for disclosure. The evidence also implies that data inputs (cookie-collected data) and internal systems (data analytic technology) complement each other and jointly affect internal information quality. In addition, the tests also help rule out the alternative explanation that consumer-facing firms simply have both more cookies and more voluntary disclosures.

Table 9

The CCPA and voluntary disclosure.

This table provides the difference-in-differences regression results based on the California Consumer Privacy Act (CCPA). The regressions are based on a sample of firms headquartered in California and uses customer-oriented (non-customer-oriented) California firms as treatment (control) firms. The dependent variable is the frequency of management sales forecasts (*SF Freq*). Control variables are included for columns (2) to (4) and are as defined in [Appendix C](#). Year fixed effects are included for columns (3) and (4). Industry and firm fixed effects are included in columns (3) and (4), respectively. All variables are winsorized at the 1st and the 99th percentiles. *t*-statistics are reported in parentheses and are based on standard errors clustered by firm. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

	SF Freq			
	(1)	(2)	(3)	(4)
Customer × Post CCPA	−0.357** (−2.51)	−0.461*** (−3.31)	−0.282** (−1.99)	−0.317*** (−2.74)
Post CCPA	−0.022 (−0.20)	0.089 (0.78)		
Customer	1.221*** (5.81)	1.734*** (6.44)		
Observations	2679	2679	2679	2679
Adjusted <i>R</i> ²	0.038	0.321	0.425	0.767
Firm Characteristics Variables	No	Yes	Yes	Yes
Industry FE	No	No	Yes	No
Firm FE	No	No	No	Yes
Year FE	No	No	Yes	Yes

Table 10

Cookie banners and the usefulness of cookies in voluntary disclosure.

This table tabulates the regression results regarding the effect of cookie consent banners on the usefulness of cookies in voluntary disclosure. *Disclosure Only* takes one if a cookie banner shows on a firm's website disclosing the usage of cookies and zero otherwise. *Disclosure & Opt-Out* takes one if a cookie consent banner not only discloses the usage of cookies but also allows a one-click opt-out for unnecessary cookies. the dependent variables are the frequency of management sales forecasts (*SF Freq*), the indicator for issuing sales forecasts (*SF*), FinBERT-based measure for customer-related disclosure (*Customer Disc.*), and aggregated FinBERT-based measure for qualitative disclosure about customers, marketing, and products (*Aggregated Disc.*). Control variables are included for all columns, and they are as defined in [Appendix C](#). When including data analytic technology variables, I include these variables with missing values imputed as zero as well as indicators for variables being missing. *t*-statistics are reported in parentheses and are based on standard errors clustered at the industry level. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

	SF Freq	SF	Customer Disc.	Aggregated Disc.
	(1)	(2)	(3)	(4)
Disclosure & Opt-Out × Cookie	0.005 (0.08)	0.010 (0.71)	−0.284** (−2.20)	−0.617*** (−3.35)
Disclosure Only × Cookie	−0.111 (−0.89)	−0.044* (−1.75)	−0.379** (−2.79)	−0.806*** (−3.90)
Cookie	0.138*** (3.26)	0.037** (2.60)	0.805*** (5.62)	1.300*** (5.42)
Disclosure & Opt-Out	0.502* (2.04)	0.117* (1.99)	1.095*** (3.70)	2.675*** (5.99)
Disclosure Only	0.786** (2.43)	0.238*** (3.76)	1.182*** (3.34)	2.893*** (4.59)
Observations	2860	2860	2119	2119
Adjusted <i>R</i> ²	0.238	0.251	0.243	0.305
Firm Characteristics Variables	Yes	Yes	Yes	Yes
Data Analytic Variables	Yes	Yes	Yes	Yes

6.5. The role of industry competition

In [Table 8](#), I assess whether proprietary costs moderate the effect of cookie-collected data on voluntary disclosure. I start with the GICS four-digit industries (25 in total) and define the five industries with the lowest HHI as the most competitive industries: Capital Goods (GICS: 2010), Materials (GICS: 1510), Commercial and Professional Services (GICS: 2020), Real Estate (GICS: 6010), and Consumer Services (GICS: 2530). In columns (1) and (2), I do not find significant results for highly competitive industries. In contrast, the relation remains positive and significant in less competitive industries in columns (3) and (4). Overall, the results indicate that the proprietary nature of cookie-collected data may prevent managers from disclosing information learned through cookies.

Table 11

The quality of management sales forecasts.

This table tabulates the regression results regarding the effect of cookie-collected data on the quality of voluntary disclosure, including management sales forecast precision and accuracy. Control variables are included for all columns, and they are as defined in [Appendix C](#). When including data analytic technology variables, I include these variables with missing values imputed as zero as well as indicators for variables being missing. All variables are winsorized at the 1st and the 99th percentiles. *t*-statistics are reported in parentheses and are based on standard errors clustered at the industry level. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

	SF Precision			SF Accuracy		
	(1)	(2)	(3)	(4)	(5)	(6)
Cookie	0.004*** (3.93)	0.002* (1.78)	0.002** (2.19)	0.001** (2.17)	0.001** (2.28)	0.001** (2.09)
Observations	826	821	821	927	924	924
Adjusted R^2	0.176	0.286	0.311	0.311	0.381	0.386
Firm Characteristic Variables	Yes	Yes	Yes	Yes	Yes	Yes
Data Analytic Variables	No	No	Yes	No	No	Yes
Industry FE	No	Yes	Yes	No	Yes	Yes

Table 12

The usefulness of cookies and firm characteristics.

This table tabulates the regression results assessing how the effects of cookie-collected data vary with firm characteristics. Panel A is to compare consumer-oriented and non-consumer-oriented industries, and Panel B is to compare large vs. small firms. The dependent variable is the frequency of management sales forecasts (*SF Freq*). Panel C tabulates the regression results comparing IT and non-IT firms. IT firms are defined as firms belonging to GICS industries with first two digits as 45. Control variables are included for all columns, and they are as defined in [Appendix C](#). For data analytic technology variables, I include the variable with missing values imputed as zero as well as indicators for variables being missing. All variables are winsorized at the 1st and the 99th percentiles. *t*-statistics are reported in parentheses and are based on standard errors clustered at the industry level. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

Panel A: Customer- vs. Non-Customer-Oriented Firms				
	Customer-Oriented		Non-Customer-Oriented	
	(1)	(2)	(3)	(4)
Cookie	0.119** (2.15)	0.111** (2.16)	0.054 (0.81)	0.023 (0.44)
Observations	1658	1658	1202	1202
Adjusted R^2	0.308	0.395	0.254	0.383
Firm Characteristics Variables	Yes	Yes	Yes	Yes
Data Analytic Variables	Yes	Yes	Yes	Yes
Industry FE	No	Yes	No	Yes
Panel B: Large vs. Small Firms				
	Size above Median		Size below Median	
	(1)	(2)	(3)	(4)
Cookie	0.060 (1.08)	0.053 (1.16)	0.156*** (2.85)	0.082* (1.83)
Observations	1430	1430	1430	1430
Adjusted R^2	0.295	0.470	0.314	0.349
Firm Characteristics Variables	Yes	Yes	Yes	Yes
Data Analytic Variables	Yes	Yes	Yes	Yes
Industry FE	No	Yes	No	Yes
Panel C: IT vs. Non-IT Sectors				
	IT Sector		Non-IT Sector	
	(1)	(2)	(3)	(4)
Cookie	0.079 (1.26)	0.074 (1.20)	0.112** (2.23)	0.084* (1.97)
Observations	342	342	2462	2462
Adjusted R^2	0.372	0.368	0.199	0.327
Firm Characteristics Variables	Yes	Yes	Yes	Yes
Data Analytic Variables	Yes	Yes	Yes	Yes
Industry FE	No	Yes	No	Yes

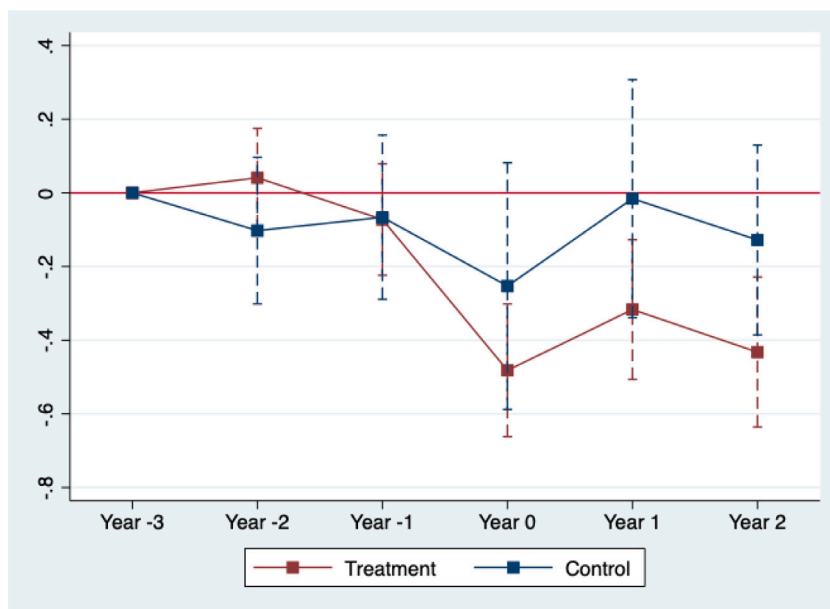


Fig. 1. Parallel Trends around the CCPA. The figures below plot the parallel trends around the California Consumer Privacy Act (CCPA). It shows the parallel trends between consumer-oriented and non-consumer-oriented California firms. Year -3 (i.e., 2017) is the benchmark year for the plot. 95% confidence intervals are plotted for each year.

6.6. DID analyses using the CCPA

To assess the effect of data privacy protection and enhance the causal evidence on the role of cookies in voluntary disclosure, I exploit the adoption of the California Consumer Privacy Act (CCPA) using a difference-in-differences design. Data privacy regulations restrict firms' ability to collect consumer data and use website cookies (e.g., Jia et al., 2021; Goldberg et al., 2024). Hence, the CCPA represents a plausibly exogenous negative shock to firms' cookie usage and data collection activities.

I validate the parallel trends assumption in Fig. 1 and tabulate the results in Table 9. Columns (1) and (2) present specifications without fixed effects. Column (3) includes industry and year fixed effects, and column (4) further includes firm fixed effects to account for time-invariant firm characteristics. The findings suggest that, relative to non-consumer-oriented California firms, consumer-oriented California firms significantly reduce the frequency of management sales forecasts after the CCPA adoption. Overall, the findings indicate that the CCPA, as a data privacy protection regulation, constrains firms' collection and usage of consumer data and thus reduces voluntary disclosure.⁴²

6.7. The role of cookie banners

In addition to the mandatory data privacy regulation, I evaluate the role of cookie consent banners as a voluntary data privacy protection mechanism. Many U.S. firms voluntarily adopt cookie banners to (1) disclose cookie usage to website users and (2) allow users to opt out of unnecessary cookies. These banners may limit firms' collection and usage of consumer data by increasing public awareness and scrutiny of such practices and restricting the volume of data collected (Goldberg et al., 2024).

I manually collect information on the presence and features of cookie consent banners from firms' websites and classify firms into three mutually exclusive groups: (1) firms without banners, (2) firms with banners with cookie usage disclosure but without opt-out options, and (3) firms with banners that provide both disclosure and opt-out options. I construct two indicator variables: *Disclosure Only*, which equals one for firms in group (2), and *Disclosure & Opt-Out*, which equals one for firms in group (3).

In Table 10, I interact these indicators with *Cookie*. The coefficient of *Cookie* captures the effect for firms without banners. The interaction term, *Disclosure Only* $\times$ *Cookie*, is negative and significant, indicating that disclosing cookie usage constrains data collection and weakens the usefulness of cookies in disclosure. However, the coefficient of *Disclosure Only & Opt-Out* $\times$ *Cookie* is insignificant, suggesting no difference relative to firms without banners. Two explanations are plausible. First, consumers' opt-out

⁴² In untabulated analyses, I also use FinBERT-based qualitative disclosure proxies as the dependent variable. However, I do not find significant results. One plausible reason is that treatment firms may have increased their discussions on consumer privacy and the CCPA in their 10-K filings after the adoption of the CCPA. Such discussions are classified as customer-related sentences by the model.

decisions may themselves be informative, allowing firms to infer consumer preferences and demographics (Johnson et al., 2020). Second, it is likely that firms with strong data analytic capabilities self-select to provide opt-out options because they are able to offset the loss of cookie-collected data.

6.8. Additional analyses and robustness checks

To enrich the analysis and validate the main inferences, I conduct several additional analyses and robustness checks.

First, Table 11 focuses on the *quality* of voluntary disclosure and replaces the dependent variables with the precision and the accuracy of management sales forecasts. Precision is defined as the negative width of range forecasts, and accuracy is the *ex post* accuracy measured as the difference between forecasted values and actual values of sales. Although the sample is smaller (available only for firms issuing sales forecasts), *Cookie* remains positively associated with both measures, suggesting that cookie-collected data enhance not only the quantity but also the quality of disclosure.

Second, in Table 12, I examine how the effect of cookie-collected data varies with firm characteristics. In line with the intuition that cookies are more useful for consumer-oriented industries, the effect is more pronounced for consumer-oriented industries. In addition, as cookies are not the only tool for firms to collect consumer data, alternative data collection tools may substitute and mitigate the effect of cookies. Ideally, I would test this prediction by combining data regarding alternative data collection tools; however, such data are extremely difficult to obtain. Therefore, I take an indirect approach by comparing the effects between large and small firms, as well as between IT and non-IT firms, in the sense that large firms (IT firms) are more likely to have alternatives than small firms (non-IT firms). Consistently, the results indicate that the effect is not significant for large firms and IT firms but only for small firms and non-IT firms, suggesting that cookies are more useful for firms with limited alternative data collection tools.

Finally, several robustness checks further substantiate the inferences of the paper. I re-estimate the regressions using alternative industry fixed effects (defined based on Fama–French 49 and SIC two-digit industries) as well as state fixed effects that capture state-level factors such as privacy regulations and legal environments (Table IA8, Internet Appendix). I also exclude firms in financial and healthcare industries, which face stricter data regulations (Table IA9). Across all specifications, the inferences remain unchanged.

7. Conclusions

Using publicly available metadata on website cookies to measure firm-collected consumer data, this study shows that consumer data collected through cookies shape firms' internal information environments and, in turn, their voluntary disclosures. Firms with more cookies issue management sales forecasts more frequently and are more likely to issue them. They also devote a larger share of qualitative disclosures in 10-K filings to customer-, marketing-, and product-related topics. Together, these findings suggest that cookie-collected consumer data enrich the internal information environment that managers rely on for financial reporting and thus facilitate external voluntary disclosure.

Building on the baseline results, I examine conditions under which cookie-collected data are more or less likely to improve internal information quality. The effects are stronger when cookies are more likely to collect data of greater relevance to firms' financial reporting or of a larger volume (e.g., first-party, analytic, and persistent cookies) and when firms have greater data analytic capabilities. I also assess managers' incentives to disclose conditional on learning incremental information from cookies. The relation between cookies and disclosure is attenuated in more competitive industries and under stronger data privacy protection mechanisms.

To provide additional evidence beyond the cross-sectional setting, I exploit the implementation of the California Consumer Privacy Act (CCPA) in a DID design. Firms more exposed to the CCPA reduce voluntary disclosure after the adoption of the CCPA, consistent with a regulatory constraint on consumer data collection weakening the informational content available for disclosure. Importantly, the DID results complement the main analyses by extending the results to panel data and strengthening the causal inference.

This study contributes to research on internal information quality and external reporting by highlighting the role of firm-collected consumer data in shaping internal information quality and external disclosures. The findings in the paper shed light on how the influx of large volumes of consumer data affects managers' acquisition and synthesis of internal information for external reporting. Methodologically, it introduces a generalizable approach to quantifying unobservable firm-collected consumer data using publicly available cookie information. The approach is generalizable and may open up new avenues for empirical work on firms' data collection practices. Finally, the findings inform ongoing debates about corporate data collection and consumer data privacy protection by showing a positive effect of consumer data on firms' information environments.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Junhao Liu reports financial support was provided by TD MDAL Research Center - University of Toronto. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Examples of cookie data and cookie consent banners

Cookie information collected by OpenWPM

As an example, I provide a snapshot of the cookie information collected by OpenWPM. As shown in the snapshot, OpenWPM collects a list of characteristics about cookies, including but not limited to cookie names, expiry, host domains, and output values.

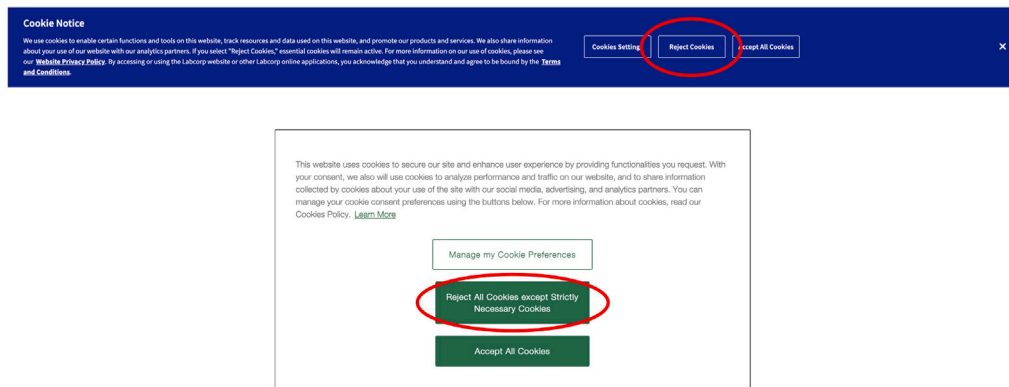
Name	Value	Domain	Path	Expires / Max-Age
id	OPT_OUT	.doubleclick.net	/	Session
DSID	AGnTgWdcJzuEBJXyMooJTlusyt2mmdWluGtvfyyY_3wR...	.doubleclick.net	/	2023-09-10T00:00:00.513Z
NID	511=i4f6iZM7gsZOnssG9GICKFqj5AbfFAwKA-dzVm9m3...	.google.ca	/	2023-12-27T20:19:33.708Z
HSID	AoM7kJONfY29hYqjt	.google.ca	/	2024-07-31T20:19:33.708Z
SSID	Add9JUJoi0Ejln1bG	.google.ca	/	2024-07-31T20:19:33.708Z
APISID	t7JBBEUesiCThwmW/AlHERwQ_UopFhQDWH	.google.ca	/	2024-07-31T20:19:33.708Z
SAPISID	ZWu4b4nRtTw9vAkH/ArzH4pWrwJ7DwlCkD	.google.ca	/	2024-07-31T20:19:33.708Z
__Secure-1PAPISID	ZWu4b4nRtTw9vAkH/ArzH4pWrwJ7DwlCkD	.google.ca	/	2024-07-31T20:19:33.708Z
__Secure-3PAPISID	ZWu4b4nRtTw9vAkH/ArzH4pWrwJ7DwlCkD	.google.ca	/	2024-07-31T20:19:33.708Z
SID	YAgSIIBESeNPdDOI_Y69YBuAVCTWDQsUPFd5Bj4o-GO...	.google.ca	/	2024-07-31T20:19:33.708Z
__Secure-1PSID	YAgSIIBESeNPdDOI_Y69YBuAVCTWDQsUPFd5Bj4o-GO...	.google.ca	/	2024-07-31T20:19:33.708Z
__Secure-3PSID	YAgSIIBESeNPdDOI_Y69YBuAVCTWDQsUPFd5Bj4o-GO...	.google.ca	/	2024-07-31T20:19:33.708Z
AEC	Ad49MVFBJHwISaMne_mm23I3ZE_YaJHaoSueg_dox_R...	.google.com	/	2023-10-01T20:19:18.860Z
1P_JAR	2023-09-03-20	.google.com	/	2023-10-03T20:31:57.879Z
SEARCH_SAMESITE	CgQlkZkB	.google.com	/	2024-02-28T20:27:52.631Z
NID	511=XfhH1fj53aHNSgei5KCRfFgOtxO8sWRHtDaqHHoJ9...	.google.com	/	2024-03-04T20:31:57.553Z
__Secure-1PSIDTS	sidts-CjIBSAxbGaFyTIFGOVpYBFOYHb_sHjTYPZ-lrHc...	.google.com	/	2024-09-02T20:08:03.186Z
__Secure-3PSIDTS	sidts-CjIBSAxbGaFyTIFGOVpYBFOYHb_sHjTYPZ-lrHc...	.google.com	/	2024-09-02T20:08:03.186Z
SIDCC	APoG2W_30D1mNgB_KWAqllQRT7_F4CG9iz9PkvN4tJy...	.google.com	/	2024-09-02T20:31:58.668Z
__Secure-1PSIDCC	APoG2W_SY73ek29ISdgPQcnzQnVwVMxaqrtnNoSeW6...	.google.com	/	2024-09-02T20:31:58.668Z
__Secure-3PSIDCC	APoG2W8nhEyk5sLO2WpMF8-PpetAPUusySwsiywWgA...	.google.com	/	2024-09-02T20:33:08.688Z
HSID	A-MI-ORjWKPNqx1bd	.google.com	/	2024-09-30T21:14:03.575Z
SSID	Ach3ISOp-m4ZyivCg	.google.com	/	2024-09-30T21:14:03.575Z
APISID	t7JBBEUesiCThwmW/AlHERwQ_UopFhQDWH	.google.com	/	2024-09-30T21:14:03.575Z
SAPISID	ZWu4b4nRtTw9vAkH/ArzH4pWrwJ7DwlCkD	.google.com	/	2024-09-30T21:14:03.575Z
__Secure-1PAPISID	ZWu4b4nRtTw9vAkH/ArzH4pWrwJ7DwlCkD	.google.com	/	2024-09-30T21:14:03.575Z
__Secure-3PAPISID	ZWu4b4nRtTw9vAkH/ArzH4pWrwJ7DwlCkD	.google.com	/	2024-09-30T21:14:03.575Z
SID	aQgSINX8PSaOWr91a-_iS9Tx-n80uu3g7pvccrct2U7NNt...	.google.com	/	2024-09-30T21:14:03.575Z
__Secure-1PSID	aQgSINX8PSaOWr91a-_iS9Tx-n80uu3g7pvccrct2U7NNt...	.google.com	/	2024-09-30T21:14:03.575Z
__Secure-3PSID	aQgSINX8PSaOWr91a-_iS9Tx-n80uu3g7pvccrct2U7NNt...	.google.com	/	2024-09-30T21:14:03.575Z
ar_debug	1	.googleadserv...	/	2023-11-25T21:25:04.781Z
TS0122c9b6	011f37387c39935364c42a45d1a14aac9f14dee7ca2d5cc...	.www.zara.com	/	Session
rid	3ad1f8c5-fa59-4e94-8877-404d123c16c0	.www.zara.com	/	2024-06-13T18:19:34.000Z
OptanonAlertBoxClosed	2023-09-03T20:32:54.119Z	.www.zara.com	/	2024-09-02T20:32:54.000Z
OptanonConsent	isGpcEnabled=0&datestamp=Sun+Sep+03+2023+16%3...	.www.zara.com	/	2024-09-02T20:33:08.000Z
cart-was-updated-in-standard	true	.zara.com	/	Session
_gat_UA-18083935-1	1	.zara.com	/	2023-09-03T20:34:08.000Z
ITXSESSIONID	9536b6335bd1f08867397552962c1f75	.zara.com	/	2023-09-03T21:03:08.800Z

First- and Third-Party cookies

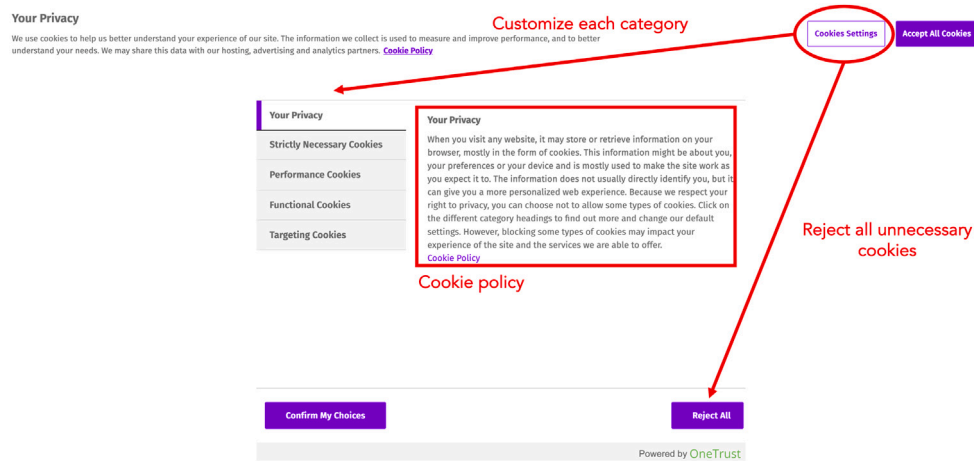
I collect the cookies from the homepage of Zara (<https://www.zara.com/>). As shown above, cookies from the domain, “.zara.com” and “.www.zara.com”, are first-party cookies. This is because the domain of cookies are the same as the website cookies. Meanwhile, Zara also employ third-party cookies on their website, as some cookies are set by Google (“.google.com”) and Adobe Doubleclick (“.doubleclick.net”).

Cookie consent banners

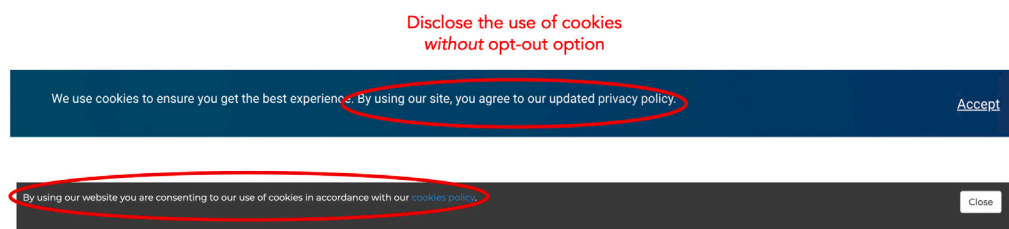
Banners with the “Reject All” option



Banners without easily accessible “Reject All” option but with customizable settings



Banners only with disclosure



Appendix B. FinBERT-Based qualitative disclosure measures

To measure firms' qualitative disclosure about customers and sales operations, I employ a fine-tuned FinBERT model to classify sentences in Items 1 (Business), 1A (Risk Factors), and 7 (Management's Discussion and Analysis) of 10-K filings into (1) customer-related, (2) marketing-related, and (3) product/service-related categories. This appendix describes the model, the fine-tuning procedure, and the construction of disclosure proxies.

FinBERT model

FinBERT is a large language model specializing in financial texts (Yang et al., 2020; Huang et al., 2023). It is pre-trained using corporate filings, analyst reports, and earnings call transcripts, making it well suited for analyzing financial disclosures.

Fine-Tune FinBERT model

I fine-tune the pre-trained FinBERT model from Huang et al. (2023) for the topic modeling task.⁴³ Below are the detailed procedures:

(1) **Training Sample Construction.** I randomly select 1500 sentences from Items 1, 1 A, and 7 of 10-K filings between 2017 and 2021 (500 sentences from each item). Sentences are required to contain at least 10 words and no more than 100 words.

(2) **Manual Labeling.** Each sentence is manually classified into one of four categories: (1) customer-related, (2) marketing-related, (3) product-related, or (4) none of the above. If a sentence relate to multiple topics, I assign it to the most relevant category to simplify model training. Among the 1500 sentences, 62 (4.13%) are labeled as customer-related, 74 (4.93%) as marketing-related, and 56 (3.73%) as product-related. Below are the guidelines for sentence classification:

- A sentence is classified as **customer-related** if it primarily discusses customer relationships, customer characteristics, customer retention or attrition, customer satisfaction, and/or interactions with existing or potential customers.
- A sentence is classified as **marketing-related** if it primarily discusses marketing strategies, advertising activities, branding, pricing, promotions, or selling channels.
- A sentence is classified as **product-related** if it primarily discusses product features, product quality, product demand, product innovation, or the lifecycle of products or services.
- A sentence is classified as **other** if it does not fall into any of the categories above.

(3) **Model Fine-Tuning.** The labeled sentences are used to fine-tune FinBERT to classify sentences into the four categories. Training follows the authors' recommended settings.⁴⁴ The fine-tuned model achieves an in-sample validation accuracy of 0.881.

(4) **Application to 10-K filings.** I apply the fine-tuned model to all sentences in Items 1, 1 A, and 7 of 10-K filings to classify them into customer-, marketing-, and product-related categories. Below are some examples:

- Customer-related:
 - We are not dependent upon any single customer, and no customer contributed 10% or more of our consolidated revenues for 2022.
 - As a result, in 2022 about 70% of our revenue was direct, which includes TI.com, as customers valued the convenience of purchasing online.
 - We also utilize proprietary tools that leverage real-time customer data to guide and personalize the customer experience.
 - Our location-specific surveys ask customers to evaluate their overall satisfaction with their rental experience and the likelihood that they will recommend our brands, as well as key elements of the rental experience.
 - A large percentage of sales at mature dealerships are made to repeat customers, and the Company estimates an additional 10% to 15% of sales result from customer referrals.
- Marketing-related:
 - Sales in this channel are driven on the basis of price, product quality, selection, and the speed and reliability of delivery.
 - During Fiscal 2022, 80% of our consolidated net sales were through our direct to consumer channels of distribution, which consist of our brand specific full-price retail stores, e-commerce websites and outlets, as well as our Tommy Bahama food and beverage operations.
 - In addition, at higher price points aimed at larger marketers (\$50M+ annual revenue), there are several large customer data platform competitors that attempt to unify many disparate sources of data to improve omnichannel advertising outcomes.
 - We expect the market will continue to shift from traditional linear video to OTT offerings.
 - We have also made significant investments in building our field commercial team and intend to make significant investments in sales and marketing efforts in the future, including initiatives to drive awareness and expand our referral channel to increase the number of drug-resistant epilepsy patients referred to Level 4 CECs.
- Product-related:
 - In addition to competing for market share with other IO products in approved indications such as lung cancer and melanoma, we face increased competition from existing competing IO products that receive FDA approval for additional indications and for new IO agents that receive FDA approval and enter the market.
 - The products and services are primarily distributed through a direct sales force and, to a lesser extent, third-party distributors.
 - The IAM-F platform allows demand side platforms (“DSPs”), advertisers, agencies, and publishers to buy and sell digital ad impressions, primarily through programmatic, real-time bidding auctions and, in some cases, through direct-bought/sold advertiser budgets.
 - To precisely create these challenging structures, our Flex product family offers differentiated technologies and application-focused capabilities for critical dielectric etch applications.

⁴³ I thank the authors for making the pre-trained FinBERT model publicly available at <https://huggingface.co/yiyanghkust/finbert-pretrain>.

⁴⁴ See <https://github.com/yya518/FinBERT/blob/master/finetune.ipynb>.

- Consumer awareness of diagnostic imaging as a less invasive and preventive screening method has added to the growth in diagnostic imaging procedures.

Disclosure proxies

I construct qualitative disclosure proxies for customers, marketing, and products (*Customer, Disc.*, *Marketing, Disc.*, and *Product, Disc.*) by scaling the number of sentences in each category by the total number of sentences in Items 1, 1A, and 7 of the corresponding 10-K filing. Because the three topics are not mutually exclusive, I also construct an aggregated measure (*Aggregated, Disc.*) by summing the three proxies to capture overall disclosure related to customers, marketing, and products.

Appendix C. Variable definitions

Variables	Definitions
<i>Voluntary Disclosure Proxies</i>	
SF Freq	The number of management sales forecasts issued.
SF	An indicator that equals one if the company issues at least one management sales forecast in a given year.
Customer Disc	The percentage of sentences pertaining to customer-related issues out of all sentences in Item 1, 1A, and 7 in 10-K filings. The topic classification is based on the fine-tuned FinBERT model as discussed in Section 4 and Appendix B.
Marketing Disc	The percentage of sentences pertaining to marketing-related issues out of all sentences in Item 1, 1A, and 7 in 10-K filings. The topic classification is based on the fine-tuned FinBERT model as discussed in Section 4 and Appendix B.
Product Disc	The percentage of sentences pertaining to product-related issues out of all sentences in Item 1, 1A, and 7 in 10-K filings. The topic classification is based on the fine-tuned FinBERT model as discussed in Section 4 and Appendix B.
Aggregated Disc	The sum of <i>Customer Disc</i> , <i>Marketing Disc</i> , and <i>Product Disc</i> .
SF Precision	The precision of management sales forecasts, defined as the negative width of forecasted range scaled by the lower bound of the range forecasts. The variable is only available for range forecasts.
SF Accuracy	Accuracy of management sales forecasts, defined as the negative unsigned forecast errors of management sales forecasts. Forecast errors are calculated as the difference between forecast and actual values of sales, scaled by market capitalization. For range forecasts, forecast value is the mid point of the range.
<i>Cookie Variables</i>	
Cookie	The natural logarithm of the total number of cookies on a company's website.
Session Cookie	The natural logarithm of the number of session cookies on a company's website. Session cookies are defined as cookies that expire within 24 h after they are sent to users' browsers.
Persistent Cookie	The natural logarithm of the number of persistent cookies on a company's website. Persistent cookies are defined as cookies that last for more than 24 h.
FP Cookie	The natural logarithm of the number of first-party cookies on a company's website. First-party cookies are defined as cookies whose domain has a Levenshtein similarity of at least 0.8 compared with the website's domain.
TP Cookie	The natural logarithm of the number of third-party cookies on a company's website. Third-party cookies are defined as cookies whose domain has a Levenshtein similarity of lower than 0.8 compared with the website's domain.
Necessary Cookie	The natural logarithm of the number of strictly necessary cookies on a company's website. Strictly necessary cookies are defined as per the GDPR, and the data are obtained from Cookiepedia and companies' cookie policy statements.
Functional Cookie	The natural logarithm of the number of functional cookies on a company's website. Functional cookies are defined as per the GDPR, and the data are obtained from Cookiepedia and companies' cookie policy statements.
Analytic Cookie	The natural logarithm of the number of analytic/performance/statistic cookies on a company's website. Analytic/performance/statistic cookies are defined as per the GDPR, and the data are obtained from Cookiepedia and companies' cookie policy statements.

Variables	Definitions
Advertising Cookie	The natural logarithm of the number of advertising/targeting cookies on a company's website. Advertising/targeting cookies are defined as per the GDPR, and the data are obtained from Cookiepedia and companies' cookie policy statements.
<i>Firm Characteristics</i>	
Size	Firm size, defined as the natural logarithm of total assets.
MB	Market-to-book ratio, calculated as: $\frac{\text{Market value of equity} + \text{Total debt}(dlc + dlrt)}{\text{Total assets}(at)}$
Leverage	Leverage ratio, calculated as: $\frac{\text{Total debt}(dlc + dlrt)}{\text{Total assets}(at)}$
ROE	Return on equity, calculated as $\frac{\text{Net income}(ni)}{\text{Stockholders' equity}(seq)}$
Loss	Loss indicator, which equals one if earnings are non-negative and zero otherwise.
Cash	Cash holding, calculated as: $\frac{\text{Cash and cash equivalents}(che)}{\text{Total assets}(at)}$
Capex	Capital expenditures, calculated as: $\frac{\text{Capital expenditures}(capx)}{\text{Net value of PP\&E}(ppent)}$
SG&A	Selling, general, and administrative expenses, calculated as (missing values in <i>xsga</i> are filled as zero): $\frac{\text{SG\&A expenses}(xsga)}{\text{Total assets}(at)}$
PPE	Property, plant, and equipment, calculated as: $\frac{\text{Net value of PP\&E}(ppent)}{\text{Total assets}(at)}$
Return	Annual stock returns during the fiscal year.
Volatility	Monthly stock return volatility, calculated as the standard deviation of monthly stock returns during the fiscal year.
Report Lag	The number of days between fiscal year ends and actual 10-K report dates.
Restate	An indicator variable that equals one if the financial statement of a given firm-year is restated and zero otherwise.
ICW	The number of internal control weaknesses in a given year.
Customer	Customer-oriented industry, takes one for industry groups under "Consumer Discretionary", "Consumer Staples", "Health Care", and "Communication Services" sectors, together with "Transportation" (2030), "Insurance" (4030), "Software and Services" (4510), and "Technology Hardware and Equipment" (4520) industry groups, as customer-oriented industries.
HHI	Herfindahl–Hirschman index for sales revenue, defined as the sum of squared market share of all firms within each GICS industry groups.
Analyst	Analyst following, defined as the natural logarithm of the number of individual analysts issuing forecasts to a company in a given year.
Inst	Institutional ownership, defined as the number of shares held by institutional investors scaled by the total number of shares.
<i>Data Analytic Variables</i>	
Data Storage	The capacity of data storage, measured as the storage space in terabyte (TB) scaled by net value of PP&E.
Server	The number of servers possessed by a firm scaled by net value of PP&E.
DBMS	The rating (0–5) for firms likelihood of using or purchasing database management software.
ERP	The rating (0–5) for firms likelihood of using or purchasing enterprise-resource-planning software.
SaaS	The rating (0–5) for firms likelihood of using or purchasing software-as-a-service solutions.
IT Employee	The percentage of IT Employees, defined as the number of IT employees scaled by the total number of employees.
IT Budget	IT budget reported to Aberdeen scaled by sales revenue.

Appendix D. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jacceco.2026.101885>.

References

- Armstrong, C.S., Konchitchki, Y., Zhang, B., 2025. Digital traffic, financial performance, and stock valuation. *Account. Rev.* 100 (6), 29–60.
- Beyer, A., Cohen, D.A., Lys, T.Z., Walther, B.R., 2010. The financial reporting environment: Review of the recent literature. *J. Account. Econ.* 50 (2–3), 296–343.
- Bhojraj, S., Lee, C.M., Oler, D.K., 2003. What's my line? A comparison of industry classification schemes for capital market research. *J. Account. Res.* 41 (5), 745–774.
- Bian, B., Ma, X., Tang, H., 2025. The supply and demand for data privacy: Evidence from mobile apps. *Working Paper*.

- Bloom, N., Sadun, R., Reenen, J.V., 2012. Americans do IT better: US multinationals and the productivity miracle. *Am. Econ. Rev.* 102 (1), 167–201.
- Bollinger, D., Kubicek, K., Cotrini, C., Basin, D., 2022. Automating cookie consent and GDPR violation detection. In: 31st USENIX Security Symposium (USENIX Security 22). pp. 2893–2910.
- Brynjolfsson, E., McElheran, K., 2016. The rapid adoption of data-driven decision-making. *Am. Econ. Rev.* 106 (5), 133–139.
- Cheng, S., Lin, Y., Lu, R., Zhang, X., 2022. Third-party cookies, data sharing, and return comovement. Working Paper.
- Cohen, M.C., 2018. Big data and service operations. *Prod. Oper. Manage.* 27 (9), 1709–1723.
- Cohn, J.B., Liu, Z., Wardlaw, M.I., 2022. Count (and count-like) data in finance. *J. Financ. Econ.* 146 (2), 529–551.
- Darrough, M.N., Stoughton, N.M., 1990. Financial disclosure policy in an entry game. *J. Account. Econ.* 12 (1–3), 219–243.
- Donders, A.R.T., Van Der Heijden, G.J., Stijnen, T., Moons, K.G., 2006. A gentle introduction to imputation of missing values. *J. Clin. Epidemiol.* 59 (10), 1087–1091.
- Dorantes, C.-A., Li, C., Peters, G.F., Richardson, V.J., 2013. The effect of enterprise systems implementation on the firm information environment. *Contemp. Account. Res.* 30 (4), 1427–1461.
- Dye, R.A., 1985. Disclosure of nonproprietary information. *J. Account. Res.* 123–145.
- Englehardt, S., 2018. Automated Discovery of Privacy Violations on the Web (Ph.D. thesis). Princeton University.
- Englehardt, S., Narayanan, A., 2016. Online tracking: A 1-million-site measurement and analysis. In: Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security. pp. 1388–1401.
- Feng, M., Li, C., McVay, S., 2009. Internal control and management guidance. *J. Account. Econ.* 48 (2–3), 190–209.
- Ferracuti, E., Koo, M., Lee, M., Stubben, S., 2024. Acquisition of customer information and corporate decision making. Working Paper.
- Forman, C., Goldfarb, A., Greenstein, S., 2005. How did location affect adoption of the commercial internet? Global village vs. urban leadership. *J. Urban Econ.* 58 (3), 389–420.
- Gallimore, J., Labro, E., 2015. The importance of the internal information environment for tax avoidance. *J. Account. Econ.* 60, 149–167.
- Goldberg, S., Johnson, G., Shriver, S., 2024. Regulating privacy online: An economic evaluation of the GDPR. *Am. Econ. J.: Econ. Policy* 16 (1), 325–358.
- Goldfarb, A., Que, V.F., 2023. The economics of digital privacy. *Annu. Rev. Econ.* 15 (1), 267–286.
- Goldfarb, A., Tucker, C., 2011a. Online display advertising: Targeting and obtrusiveness. *Mark. Sci.* 30 (3), 389–404.
- Goldfarb, A., Tucker, C., 2011b. Privacy regulation and online advertising. *Manag. Sci.* 57, 57–71.
- Goldfarb, A., Tucker, C., 2012. Shifts in privacy concerns. *Am. Econ. Rev.* 102, 349–353.
- Goldfarb, A., Tucker, C., 2019. Digital economics. *J. Econ. Lit.* 57 (1), 3–43.
- Hemmer, T., Labro, E., 2008. On the optimal relation between the properties of managerial and financial reporting systems. *J. Account. Res.* 46 (5), 1209–1240.
- Huang, A.H., Wang, H., Yang, Y., 2023. FinBERT: A large language model for extracting information from financial text. *Contemp. Account. Res.* 40 (2), 806–841.
- Ittner, C.D., Michels, J., 2017. Risk-based forecasting and planning and management earnings forecasts. *Rev. Account. Stud.* 22, 1005–1047.
- Jia, J., Jin, G.Z., Wagman, L., 2021. The short-run effects of the general data protection regulation on technology venture investment. *Mark. Sci.* 40 (4), 661–684.
- Johnson, G.A., Shriver, S.K., Du, S., 2020. Consumer privacy choice in online advertising: Who opts out and at what cost to industry? *Mark. Sci.* 39 (1), 33–51.
- Jung, W.-O., Kwon, Y.K., 1988. Disclosure when the market is unsure of information endowment of managers. *J. Account. Res.* 146–153.
- Kepler, J.D., McClure, C., Stewart, C.R., 2025. Quantifying intangible capital shares and their implications for productivity and growth. Working Paper.
- Kiron, D., Ferguson, R.B., Prentice, P.K., 2013. From value to vision: Reimagining the possible with data analytics. *MIT Sloan Manag. Rev.* 54 (3), 1.
- Lambrecht, A., Tucker, C., 2013. When does retargeting work? Information specificity in online advertising. *J. Mark. Res.* 50 (5), 561–576.
- Larcker, D.F., Rusticus, T.O., 2010. On the use of instrumental variables in accounting research. *J. Account. Econ.* 49 (3), 186–205.
- LaValle, S., Lesser, E., Shockley, R., Hopkins, M.S., Kruschwitz, N., 2010. Big data, analytics and the path from insights to value. *MIT Sloan Manag. Rev.*
- Levenshtein, V.I., 1966. Binary codes capable of correcting deletions, insertions, and reversals. In: *Soviet Physics Doklady*. vol. 10, Soviet Union, pp. 707–710.
- Martin, K.D., Murphy, P.E., 2017. The role of data privacy in marketing. *J. Acad. Mark. Sci.* 45 (2), 135–155.
- Munir, S., Siby, S., Iqbal, U., Englehardt, S., Shafiq, Z., Troncoso, C., 2023. Cookiegraph: Understanding and detecting first-party tracking cookies. In: Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security. pp. 3490–3504.
- Neely, M.P., Cook, J.S., 2011. Fifteen years of data and information quality literature: Developing a research agenda for accounting. *J. Inf. Syst.* 25 (1), 79–108.
- Samuels, D., 2021. Government procurement and changes in firm transparency. *Account. Rev.* 96, 401–430.
- Tong, S., Luo, X., Xu, B., 2020. Personalized mobile marketing strategies. *J. Acad. Mark. Sci.* 48 (1), 64–78.
- Veldkamp, L., 2023. Valuing data as an asset. *Rev. Financ.* 27 (5), 1545–1562.
- Verrecchia, R.E., 1983. Discretionary disclosure. *J. Account. Econ.* 5, 179–194.
- Verrecchia, R.E., 1990. Information quality and discretionary disclosure. *J. Account. Econ.* 12, 365–380.
- Wang, H., Lee, M.K., Wang, C., 1998. Consumer privacy concerns about internet marketing. *Commun. ACM* 41 (3), 63–70.
- Wedel, M., Kannan, P., 2016. Marketing analytics for data-rich environments. *J. Mark.* 80 (6), 97–121.
- Yang, Y., Uy, M.C.S., Huang, A., 2020. Finbert: A pretrained language model for financial communications. *arXiv preprint arXiv:2006.08097*.